Residues and Duality

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May 1966

Preface

In the spring of 1963 I suggested to Grothendieck the possibility of my running a seminar at Harvard on his theory of duality for coherent sheaves — a theory which had been hinted at in his talk to the Séminaire Bourbaki in 1957 [Gro57b], and in his talk to the International Congress of Mathematicians in 1958 [Gro60b], but had never been developed systematically. He agreed, saying that he would provide an outline of the material, if I would fill in the details and write up lecture notes of the seminar. During the summer of 1963, he wrote a series of "pré-notes" [Gro63] which were to be the basis for the seminar.

I quote from the preface of the pré-notes:

> Les présentes notes donnent une esquisse assez détaillée d'une théorie cohomologique de la dualité des Modules cohérents sur les préschémas. Les idées principales de la théorie m'étaient connues dès 1959, mais le manque de fondements adéquats d'Algèbre Homologique m'avait empêché d'aborder une rédaction d'ensemble. Cette lacune de fondements est sur le point d'être comblée par la thèse de Verdier, ce qui rend en principe possible un exposé satisfaisant. Il est d'ailleurs apparu depuis qu'il existe des théories cohomologiques de dualité formellement très analogues à celle développée ici dans toutes sortes d'autres contextes: faisceaux cohérents sur les espaces analytiques, faisceaux abéliens sur les espaces topologiques (Verdier), modules galoisiens (Verdier, Tate), faisceaux de torsion sur les schémas munis de leur topologie étale, corps de classe en tous genres... Cela me semble une raison assez sérieuse pour se familiariser avec le yoga général de la dualité dans un cas type, comme la théorie cohomologique des résidus.
>
> La théorie consiste pour l'essentiel dans des questions de variance: construction d'un foncteur $f^!$ et d'un homomorphisme-trace $\underline{\underline{R}}f_* f^! \to \mathrm{id}$. La construction donnée ici est compliquée et indirecte et n'est pas valable sous des conditions aussi générales qu'on est en droit de s'y attendre.

> Il faudra sans doute une idée nouvelle pour apporter des simplifications substantielles.

The seminar took place in the fall and winter of 1963–64, with the assistance of David Mumford, John Tate, Stephen Lichtenbaum, John Fogarty, and others, and gave rise to a series of six exposés which were circulated to a limited audience under the title "Séminaire Hartshorne". The present notes are a revised, expanded, and completed version of the previous notes.

I would like to take this opportunity to thank all those people who have helped in the course of this work, and in particular A. Grothendieck, who gave continual support and encouragement throughout the whole project.

R.H.
Cambridge, May 1966

Contents

Chapter 1

Introduction

The way up and the way down are one and the same.

–Heraclitus

The main purpose of these notes is to prove a duality theorem for cohomology of quasi-coherent sheaves, with respect to a proper morphism of locally noetherian preschemes. Various such theorems are already known. Typical is the duality theorem for a non-singular complete curve X over an algebraically closed field k, which says that

$$h^0(D) = h^1(K - D),$$

where D is a divisor, K is the canonical divisor, and

$$h^i(D) = \dim_k \mathrm{H}^i(X, \mathcal{O}_X(D))$$

for any i, and any divisor D. (See e.g. [**Ch. II**; Ser59] for a proof.)

Various attempts were made to generalize this theorem to varieties of higher dimension, and as Zariski points out in his report [Zar56], his generalization of a lemma of Enriques–Severi [Zar52] is equivalent to the statement that for a normal projective variety X of dimension n over k, for any divisor D,

$$h^0(D) = h^n(K - D).$$

This is also equivalent to a theorem of Serre [Ser55a] §76, Thm. 4 on the vanishing of the cohomology group $\mathrm{H}^1(X, \mathcal{O}_X(-m))$ for m large and $\mathcal{L}$ locally free. Using a related theorem [Ser55a] §75, Thm. 3, Zariski shows how one can deduce on a non-singular projective variety the formula

$$h^i(D) = h^{n-i}(K - D)$$

for $0 \le i \le n$. In terms of sheaves, this result corresponds to the fact that the k-vector spaces

$$\mathrm{H}^i(X, \mathcal{F}) \quad \text{and} \quad \mathrm{H}^{n-i}\big(X, \mathcal{F}^\vee \otimes \omega\big)$$

are dual to each other, where $\mathcal{F}$ is a locally free sheaf, $\mathcal{F}^\vee$ is the dual sheaf $\underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{O}_X)$, and $\omega = \Omega^n_{X/k}$ is the sheaf of n-differentials on X. Serre [Ser55b] gives a proof of this same theorem by analytic methods for a compact complex analytic manifold X.

Grothendieck [Gro57b] gave some generalizations of these theorems for non-singular projective varieties and then in [Gro60b] announced the general theorem for schemes proper over a field, with arbitrary singularities, which is the subject of the present lecture notes.

To motivate the statement of our main theorem, let us consider the case of projective space $X = \mathbb{P}^n_k$ over an algebraically closed field k. Then there is a canonical isomorphism

$$\mathrm{H}^n(X, \omega) \cong k \tag{1.1}$$

where $\omega = \Omega^n_{X/k}$ is the sheaf of n-differentials. Combining this with the Yoneda pairing

$$\mathrm{H}^i(X, \mathcal{F}) \times \mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \to \mathrm{H}^n(X, \omega) \tag{1.2}$$

we obtain a pairing

$$\mathrm{H}^i(X, \mathcal{F}) \times \mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \to k \tag{1.3}$$

which one shows easily to be a perfect pairing [**SGA 62**; **expose 12**]. This generalizes the statements above, because for a locally free sheaf $\mathcal{F}$,

$$\mathrm{Ext}^{n-i}(\mathcal{F}, \omega) = \mathrm{Ext}^{n-i}\big(\mathcal{O}_X, \mathcal{F}^\vee \otimes \omega\big) = \mathrm{H}^{n-i}\big(X, \mathcal{F}^\vee \otimes \omega\big).$$

Another way of looking at our duality pairing is as an isomorphism

$$\mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \cong \mathrm{Hom}_k\big(\mathrm{H}^i(X, \mathcal{F}), k\big). \tag{1.4}$$

Since everything is linear over k, we may introduce a k-vector space G, and have an isomorphism

$$\mathrm{Ext}^{n-i}_X\big(\mathcal{F}, G \otimes_k \omega\big) \to \mathrm{Hom}_k\big(\mathrm{H}^i(X, \mathcal{F}), G\big). \tag{1.5}$$

Before proceeding further, we must introduce the derived category. It will be discussed in detail in Chapter I, but for the moment it will be sufficient to know the following: For each abelian category $\mathcal{A}$, there is a category $\mathrm{D}(\mathcal{A})$, called the derived category of $\mathcal{A}$, whose objects are complexes of objects of $\mathcal{A}$. If $F\colon \mathcal{A} \to \mathcal{B}$ is an additive functor from one abelian category to another, then under reasonable conditions there is a right derived functor $\underline{\underline{\mathrm{R}}}F\colon \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ with the property that for any $X \in \mathrm{Ob}\,\mathcal{A}$, if X denotes also the complex which is X in degree zero, and zero elsewhere, then

$$\mathrm{H}^i(\underline{\underline{\mathrm{R}}}F(X)) = R^iF(X),$$

where R^iF is the ordinary i^{th} right derived functor of F. Finally, if $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ are two functors then

$$\underline{\underline{\mathrm{R}}}(G \circ F) = \underline{\underline{\mathrm{R}}}G \circ \underline{\underline{\mathrm{R}}}F.$$

This replaces the old-fashioned spectral sequence of a composite functor.

Now we can jazz up our duality for projective space as follows. We replace k by a prescheme Y, so that $X = \mathbb{P}^n_Y$. We consider the derived categories $\mathrm{D}(X)$ and $\mathrm{D}(Y)$ of the categories of $\mathcal{O}_X$-modules and $\mathcal{O}_Y$-modules, respectively. Then cohomology H^i becomes $\underline{\underline{\mathrm{R}}}f_*$, the derived functor of the direct image functor f_*, where $f\colon X \to Y$ is the projection. Ext becomes the derived functor $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}$ of Hom. We define

$$f^!(\mathcal{G}) = f^*(\mathcal{G}) \otimes \omega$$

for $\mathcal{G} \in \mathrm{D}(Y)$, and we replace $\mathcal{F}$ by a complex of sheaves $\mathcal{F} \in \mathrm{D}(X)$. Then the isomorphism (1.1) gives us an isomorphism

$$\underline{\underline{\mathrm{R}}}f_*\, f^!\mathcal{G} \xrightarrow{\sim} \mathcal{G} \tag{1.6}$$

which we call the trace map.

The Yoneda pairing reappears as a natural map

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_X\big(\mathcal{F}, f^!\mathcal{G}\big) \to \underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}}f_*\mathcal{F}, \underline{\underline{\mathrm{R}}}f_*\, f^!\mathcal{G}\big), \tag{1.7}$$

which, composed with the trace map (1.6) gives us the duality morphism

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_X\big(\mathcal{F}, f^!\mathcal{G}\big) \to \underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}}f_*\mathcal{F}, \mathcal{G}\big) \tag{1.8}$$

which generalizes (1.5). This is easily proved to be an isomorphism ([III, 5.1] below) under suitable hypotheses on $Y, \mathcal{F}, \mathcal{G}$. In fact, the proof is nothing but “general nonsense” once one has the isomorphism (1.4).

Having examined the case of projective space, we can state the following ideal theorem, which is the primum mobile of these notes, although it may never appear explicitly in this form.

Ideal Theorem.

(a) For every morphism $f\colon X \to Y$ of finite type of preschemes, there is a functor $f^!$ such that

1) if $g\colon Y \to Z$ is a second morphism of finite type, then
$$(gf)^! = f^! \circ g^!;$$

2) if f is a smooth morphism, then
$$f^!(\mathcal{G}) = f^*(\mathcal{G}) \otimes \omega,$$
where $\omega = \Omega^n_{X/Y}$ is the sheaf of highest order differentials;

3) if f is a finite morphism, then
$$f^!(\mathcal{G}) = \underline{\mathrm{Hom}}_{\mathcal{O}_Y}\big(f_*\mathcal{O}_X, \mathcal{G}\big)^{\sim}.$$

(b) For every proper morphism $f\colon X \to Y$ of preschemes, there is a trace morphism
$$\mathrm{Tr}_f\colon \underline{\underline{\mathrm{R}}} f_* f^! \to \mathrm{id}$$
of functors from $\mathrm{D}(Y)$ to $\mathrm{D}(Y)$ such that

1) if $g\colon Y \to Z$ is a second proper morphism, then $\mathrm{Tr}_{g\circ f} = \mathrm{Tr}_g \circ \mathrm{Tr}_f$.

2) if $X = \mathbb{P}^n_Y$, then Tr_f is the map deduced from the canonical isomorphism $R^n f_*(\omega) \cong \mathcal{O}_Y$.

3) if f is a finite morphism, then Tr_f is obtained from the natural map "evaluation at one"
$$\underline{\mathrm{Hom}}_{\mathcal{O}_Y}\big(f_*\mathcal{O}_X, G\big) \to G.$$

(c) If $f\colon X \to Y$ is a proper morphism, then the duality morphism
$$\theta_f\colon \underline{\underline{\mathrm{R}}}\mathrm{Hom}_X\big(F, f^!G\big) \to \underline{\underline{\mathrm{R}}}\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}} f_*F, G\big)$$
obtained by composing the natural map (7) above with Tr_f, is an isomorphism for $F \in \mathrm{D}(X)$ and $G \in \mathrm{D}(Y)$.

It should be noted that we have deliberately left certain technical details out of the above statement, for ease of reading. Thus it seems reasonable to make these statements only for complexes of quasi-coherent sheaves, or rather for complexes of arbitrary sheaves whose cohomology sheaves are quasi-coherent. (We denote this category by $\mathrm{D}_{\mathrm{qc}}(Y)$ or $\mathrm{D}_{\mathrm{qc}}(X)$.) In fact, we give an example in the case of a finite morphism to show that the duality theorem (c) fails if G is not quasi-coherent (see example following [III, 6.7]). Secondly one must expect certain boundedness conditions on the complexes involved, namely F should be bounded above (we write $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$) and G should be bounded below ($G \in \mathrm{D}^+_{\mathrm{qc}}(Y)$) for the duality theorem. Finally, questions of variance will be very important in the proof, and so the equals signs in (a) and (b) must be taken with a grain of salt. We must preserve very carefully the distinction between "equals" and "is canonically isomorphic to". Hence we will have more precise statements below.

As to proving the ideal theorem, we succeed only partially. Certainly the conditions under which we can prove it will suffice for most applications, but it is unsatisfying to have restrictive hypotheses which are apparently not essential to the truth of the theorem. I mention four sets of hypotheses under which the theorem can be proved.

(i) For the category of noetherian preschemes of finite Krull dimension, and morphisms $f\colon X \to Y$ which can be factored through a suitable projective space $\mathbb{P}^N_Y$, we have (a1)–(a3) for $\mathrm{D}^+_{\mathrm{qc}}(Y)$ [I, 8.7], (b1)–(b3) for $\mathrm{D}^+_{\mathrm{qc}}(Y)$ [I, 10.5], and (c) for $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$ and $G \in \mathrm{D}^+_{\mathrm{qc}}(Y)$ [III, 11.1].

(ii) For the category of noetherian preschemes which admit dualizing complexes (see [V, §2]; this implies in particular that the preschemes have finite Krull dimension) and morphisms whose fibres are of bounded dimension, we have (a1)–(a3) and (b1)–(b3) for $\mathrm{D}^+_c(Y)$, and (c) for $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$, $G \in \mathrm{D}^+_c(Y)$ [VII, 3.4]. Here the subscript c denotes complexes with coherent cohomology sheaves.

(iii) For the category of noetherian preschemes of finite Krull dimension and smooth morphisms, we have (a1)–(a3) for $\mathrm{D}^+_{\mathrm{qc}}(Y)$, (b1)–(b3) for $G \in \mathrm{D}^b_{\mathrm{qc}}(Y)$, and (c) for $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$ and $G \in \mathrm{D}^b_{\mathrm{qc}}(Y)$ [VII, 4.3]. Here the exponent "b" denotes complexes which are bounded in both directions, i.e., finite.

(iv) Recently P. Deligne has shown (unpublished) that for the category of noetherian preschemes, one can construct $f^!$ and Tr_f satisfying (a), (b), and (c), working with $F \in \mathrm{D}(\mathrm{Qco}(X))$ and $G \in \mathrm{D}^+(\mathrm{Qco}(Y))$, the derived categories of the categories of quasi-coherent sheaves on X and Y, respectively.

The principal difficulty has been the lack of a suitable construction of the functor $f^!$: to define it locally, and then glue. Our procedure is to define $f^!$ for a finite or a smooth morphism, where we have it given to us by (a2) and (a3) [III, §6,§11]. Thus by composition we can obtain it for any morphism which can be factored into a finite morphism followed by a smooth morphism [III, §8]. However, the derived category is not a local object, so we cannot glue these local determinations of $f^!$ to obtain a global one. We resort to a clumsy, round-about procedure of defining $f^!$ for a special class of complexes, called residual complexes [VI, §1], and then pulling ourselves up by our bootstraps to get it for arbitrary complexes [VII, §3]. But our result has the unpleasant hypotheses of (ii) above.

Deligne's construction of $f^!$ is entirely different, and proceeds essentially by representing the functor (for $F \in \mathrm{D}(X)$)

$$F \longmapsto \underline{\underline{\mathrm{R}}}\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}}f_*F, G\big).$$

This approach has the advantage of giving the duality theorem immediately. He also has a method for calculating $f^!G$ locally on X. However, it is not immediately clear from his construction that the properties (a2), (a3) and (b2), (b3) hold. In fact, their proof will probably require some knowledge of the duality theorem as we have proved it here. At least one can hope that some combination of the two approaches will provide substantial simplifications of the theory at a later date.

A second difficulty, which lurks on the fringes of these notes, is that the derived category seems to be a little too big when it comes to unbounded complexes. For example, one can have two unequal morphisms $f, g\colon X \to Y$ in $\mathrm{D}^+(\mathcal{A})$ (where $\mathcal{A}$ is an abelian category) such that their restrictions to each truncation of the complex X to a bounded complex are equal. This gives rise to some trouble with unbounded complexes (see the problem after [II, 5.7], the boundedness hypotheses in [IV, 3.1] and [VII, 4.3a], and the remark following [VI, 1.1c]). Perhaps one will have to replace D^+ by the categories $\operatorname{ind} D^b$ and $\operatorname{pro} D^b$, which however may not be triangulated categories.

Thirdly, some discussion of our noetherian hypotheses is in order. In the present state of the theory, the noetherian hypotheses are well entrenched. We have used them in [II, §7] for the structure of the injective objects in the category of sheaves on a locally noetherian prescheme, and its consequence that $\mathrm{D}^+(\mathrm{Qco}(X)) \to D^+_{\mathrm{qc}}(X)$ is an equivalence of categories. We have used them in the construction of the trace map for projective space [III, §4], in the construction of $f^!$ for a finite morphism [III, §6], in the whole theory of dualizing complexes [V, §2], in the definition of residual complexes [VI, §1], and so forth. Our finite Krull dimension hypotheses are often needed only to make possible the definition of $\mathbb{R}f_*$ and $f^!$ for unbounded complexes — a problem which will disappear when the second difficulty above is solved, and the relation between $\mathrm{D}(\mathbf{Qco}(X))$ and $\mathrm{D}_{\mathrm{qc}}(X)$ is better understood. It is certainly clear that our methods of proof rely heavily on noetherian hypotheses. I expect, however, that once a suitable statement is obtained, e.g., in case (iii) above, one could expect to prove the theorem without noetherian hypotheses, by reducing to the noetherian case. More satisfactory, of course, would be a treatment where noetherian hypotheses on the base were eliminated from the proofs as well as from the statements. When that is achieved, it will be reasonable to state the duality theorem for a proper morphism of ringed topos, whose fibres are noetherian schemes... At the present, however, this must remain a dream of things to come.

Now we will give the reader a brief description of the organization of these notes.

Chapter I gives the language of derived categories, which is used continually in the sequel. Sections 1–5, containing the definition of derived categories and derived functors, are essential, while sections 6 and 7 are refinements which are needed in proofs later on. This chapter is a self-contained treatment of the subject, and makes no reference to algebraic geometry, so can be used independently as an introduction to the notion of derived category. Sections 1–6 (except for the notion of localizing subcategory and the corresponding categories $\mathrm{K}_{A'}(\mathcal{A})$, $\mathrm{D}_{A'}(\mathcal{A})$, etc.) are taken almost without change from notes of Verdier, and should appear in his thesis [Ver63b].

Chapter II is a fairly systematic treatment of the applications of the language of derived categories to the category of sheaves on a prescheme. We consider the functors Γ, f_*, Hom, $\underline{\mathrm{Hom}}$, $\otimes$, f^*, their derived functors, and relations between these derived functors, such as associativity formulae. There is really no new mathematics involved here, since it is merely a translation into a new language of known results. However, some care is taken to see what hypotheses are needed. Only section 7 is notable new material. Here we give the structure of injective $\mathcal{O}_X$-modules on a locally noetherian prescheme X, showing that they are all direct sums of certain indecomposable injectives $J(x, x')$, for pairs of points x specializing to x' of X. This extends results of Matlis [Mat58] and Gabriel [Gab62] for the case of quasi-coherent sheaves.

Chapter III contains everything we can say about duality for projective morphisms, and if the reader cares only for them, he may stop at the end of this chapter. However, it should be noted that the following chapters, besides giving us the tools for the proof of the general duality theorem in Chapter VII, also give much new insight into the nature of duality for a projective morphism.

There are so many situations in which we have a functor $f^!$ like the one mentioned in the ideal theorem above, that we use different notations for them. Thus we have

$f^\sharp$ for a smooth morphism in section 2, $f^\flat$ for a finite morphism in section 6, and $f^!$ for an embeddable morphism in section 8. Later we will also have f^y, f^z [VI, §2] and f^Δ [VI, §3] for residual complexes.

In sections 3, 4, and 5 we recall the explicit calculations of cohomology for projective space, and give the old duality for projective space in the new language of derived categories. Sections 6 and 7 give the corresponding formalism for finite morphisms, and its relation to the case of smooth morphisms, so that in sections 8, 10, and 11 we can prove the ideal duality theorem for projective morphisms. Section 9 gives the formalism of a residue symbol which generalizes the classical residue of a differential on a curve. Even over the complex numbers, this important concept of residue for varieties of dimension greater than one was not known before.

In Chapter IV we study "local cohomology", or cohomology with restricted supports, of abelian sheaves on a locally noetherian topological space, generalizing results of [LC, §16]. In particular, we discuss various cohomological properties which a sheaf or complex of sheaves may have with respect to certain families of supports. This gives rise to the notions of depth, Cousin complex, Cohen–Macaulay complex, and Gorenstein complex. The results of this chapter are independent of all other chapters of these notes, and so may be of use elsewhere, although their only application for the moment is to the theory of duality on preschemes.

Chapter V discusses dualizing complexes (read section 0 to find out what one is) and gives a duality theorem for modules over a local ring, generalizing results of [LC, §16]. In particular, the dualizing functor

$$\underline{\mathrm{D}} = \underline{\underline{\mathrm{R}}\mathrm{Hom}}(\,\bullet\,, R^\bullet)$$

treated here will be useful for our bootstrap operation (construction of $f^!$) later, because it interchanges $\otimes$ and $\underline{\mathrm{Hom}}$, $f^!$ and $\underline{\underline{\mathrm{L}}}f_*$, and commutes with $\underline{\underline{\mathrm{R}}}f_*$ (duality theorem).

In Chapter VI we prepare for the final duality theorem by giving the construction of $f^!$ and Tr_f for residual complexes. This is accomplished by a delicate glueing procedure which is the most difficult part of the theory, so we have given it in some detail. Perhaps some day this type of construction will be done more elegantly using the language of fibred categories and results of Giraud's thesis [Gir60].

Chapter VII contains two main results. The first is the residue theorem, which generalizes the classical theorem that the sum of the residues of a differential on a curve is zero, and which is proved by reduction to that case. The second is the proof of the duality theorem for a proper morphism, which, now that all the functorial machinery has been set up, is little more than putting together the pieces of a jigsaw puzzle.

It remains to give the reader some perspective by listing some topics which have a logical place in these notes, but which are not here.

1. The cohomology class associated to a cycle. Let X be proper and smooth of dimension n over a field k. In [8, §4] it is shown how to associate to each non-singular subvariety Y of codimension p of X, a cohomology class

$$\mathrm{P}_X(Y) \in \mathrm{H}^p\big(X, \Omega^p_{X/k}\big).$$

Now this can be done for an arbitrary subvariety of X, using the remarks in [9]. One defines

$$\omega_Y = \mathrm{H}^{-n+p}\big(g^! k\big),$$

where $g\colon Y \to k$ is the projection. Then there is a canonical element

$$\eta \in \mathrm{Ext}^{-n+p}_{\mathcal{O}_Y}\big(\omega_Y, g^! k\big).$$

One defines a natural map

$$\Omega^{n-p}_{Y/k} \to \omega_Y,$$

which, together with η and the construction of [8, §4], gives the cohomology class $\mathrm{P}_X(Y)$. One proves the fundamental theorem that formation of the cohomology class associated to a cycle takes an intersection of cycles into the cup-product of their cohomology classes.

2. The theory of Poincaré duality and the Gysin homomorphism can be developed as in [8, §4].

3. A Lefschetz–Verdier fixed point formula for coherent sheaves on a scheme proper over a field, to generalize [21, Theorem 2]. In particular, the determination of the local contribution at a non-simple fixed point presents an interesting topic for future investigation.

4. The still lacking theory of duality for complexes with differential operators as boundary operator, its ties with ordinary singular homology theory and with vector bundles with integrable connections, as suggested in [22].

A remark on references: Theorem 5.1 of Chapter III is referred to as "Theorem 5.1" in Chapter III, and as [III, 5.1] elsewhere. References to the bibliography at the end are given by square brackets with an arabic numeral or some capital letters, e.g., [14, (31.I)] or [EGA III, 2.1.12].

Chapter 2

The Derived Category

2.1 Introduction

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, and let $F\colon \mathcal{A} \to \mathcal{B}$ be a functor. Our purpose is to define, functorially for each $X \in \operatorname{Ob}\mathcal{A}$, a complex $\mathbb{R}F(X)$ whose cohomology groups are the right derived functors of F acting on X, namely $R^iF(X)$. In general, we will define $\mathbb{R}F(X^\bullet)$ for any complex $X^\bullet$ of objects of $\mathcal{A}$.

More precisely, $\mathbb{R}F$ will be a functor from the derived category $\mathcal{D}(\mathcal{A})$ of $\mathcal{A}$ to the derived category $\mathcal{D}(\mathcal{B})$ of $\mathcal{B}$.

The derived category $\mathcal{D}(\mathcal{A})$ is obtained as follows: one first considers the category $\mathcal{K}(\mathcal{A})$, whose objects are complexes of elements of $\mathcal{A}$, and whose morphisms are homotopy equivalence classes of morphisms of complexes. The category $\mathcal{D}(\mathcal{A})$ is obtained by "localizing" $\mathcal{K}(\mathcal{A})$ so that every morphism in $\mathcal{K}(\mathcal{A})$ which induces an isomorphism on cohomology becomes an isomorphism in $\mathcal{D}(\mathcal{A})$. This process of localization will be explained in general below.

Although $\mathcal{A}$ and $\mathcal{B}$ above are assumed to be abelian categories, the categories $\mathcal{K}(\mathcal{A})$, $\mathcal{D}(\mathcal{A})$, etc., are in general not abelian. However, they can be given a structure which carries enough information for our purposes, namely a structure of triangulated category. Thus we will be led to study triangulated categories.

2.2 Triangulated Categories

2.2.1 Definition

A triangulated category is an additive category $\mathcal{C}$, together with

a) an automorphism $T\colon \mathcal{C} \to \mathcal{C}$ called the *translation functor*, and

b) a collection of sextuples (X, Y, Z, u, v, w), called the *triangles* of $\mathcal{C}$,

where in each sextuple $X, Y, Z \in \operatorname{Ob} \mathcal{C}$ and

$$u\colon X \to Y, \quad v\colon Y \to Z, \quad w\colon Z \to T(X)$$

are morphisms. A triangle is usually written

$$X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} T(X).$$

A morphism of triangles

$$(X, Y, Z, u, v, w) \to (X', Y', Z', u', v', w')$$

is a commutative diagram

$$\begin{tikzcd}
X \arrow[r, "u"] \arrow[d, "f"'] & Y \arrow[r, "v"] \arrow[d, "g"'] & Z \arrow[r, "w"] \arrow[d, "h"'] & T(X) \arrow[d, "T(f)"'] \\
X' \arrow[r, "u'"] & Y' \arrow[r, "v'"] & Z' \arrow[r, "w'"] & T(X')
\end{tikzcd}$$

This data is subject to the following axioms:

(TR1) (a) Every sextuple isomorphic to a triangle is a triangle.

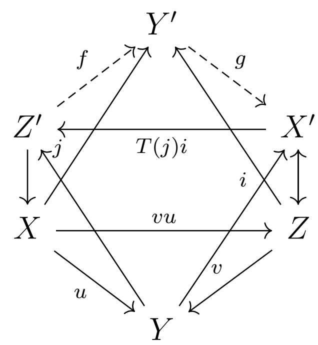
(b) Every morphism $u\colon X \to Y$ can be embedded in a triangle.

(c) The sextuple $(X, X, 0, \mathrm{id}_X, 0, 0)$ is a triangle.

(TR2) (X, Y, Z, u, v, w) is a triangle if and only if $(Y, Z, T(X), v, w, -T(u))$ is a triangle.

(TR3) Given two triangles (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w'), and morphisms $f\colon X \to X'$, $g\colon Y \to Y'$ such that $u'f = gu$, there exists a morphism $h\colon Z \to Z'$ (not necessarily unique) so that (f, g, h) is a morphism of triangles.

(TR4) (The octahedral axiom.)



Suppose given triangles

$$(X, Y, Z', u, j, \cdot),$$
$$(Y, Z, X', v, \cdot, i),$$
$$(X, Z, Y', vu, \cdot, \cdot)$$

such that

$$(Z', Y', X', f, g, T(j)i)$$

is a triangle. Then there exist morphisms $f\colon Z' \to Y'$ and $g\colon Y' \to X'$ such that the two other faces of the octahedron with f, g as edges are commutative diagrams.

Definition 1. *An additive functor $F\colon \mathcal{C} \to \mathcal{C}'$ from one triangulated category to another is called a* (covariant) ∂-functor *if it commutes with the translation functor and takes triangles into triangles. A contravariant ∂-functor takes triangles into triangles with the arrows reversed and sends the translation functor into its inverse.*

Definition 2. *An additive functor $H\colon \mathcal{C} \to \mathcal{A}$ from a triangulated category to an abelian category is called a* covariant cohomological functor *if whenever (X, Y, Z, u, v, w) is a triangle, the long sequence*

$$\cdots \to H(T^iX) \to H(T^iY) \to H(T^iZ) \to H(T^{i+1}X) \to \cdots$$

is exact (the morphisms being $H(T^iu)$, etc.). If H is a cohomological functor, we often write $H^i(X)$ for $H(T^iX)$, $i \in \mathbb{Z}$. One defines a contravariant cohomological functor by reversing the arrows.

Proposition 1. *a) The composition of any two consecutive morphisms in a triangle is zero.*

b) If $\mathcal{C}$ is a triangulated category and M an object of $\mathcal{C}$, then $\mathrm{Hom}_{\mathcal{C}}(M, \cdot)$ and $\mathrm{Hom}_{\mathcal{C}}(\cdot, M)$ are cohomological functors on $\mathcal{C}$.

c) *If in the situation of (TR3) f and g are isomorphisms, then h is also an isomorphism.*

Proof. a) Let (X, Y, Z, u, v, w) be a triangle. By (TR2) it is sufficient to show that $vu = 0$. By (TR1), $(Z, Z, 0, \mathrm{id}_Z, 0, 0)$ is a triangle. Also by (TR2), $(Y, Z, T(X), v, w, -T(u))$ is a triangle. We apply (TR3) to the maps $v\colon Y \to Z$ and $\mathrm{id}_Z\colon Z \to Z$, giving a morphism of triangles. It follows that $T(v)(-T(u)) = 0$, or since T is an automorphism, $vu = 0$.

b) Let (X, Y, Z, u, v, w) be a triangle. To show $\mathrm{Hom}_{\mathcal{C}}(M, \cdot)$ is a cohomological functor, it suffices by (TR2) to show the sequence

$$\mathrm{Hom}_{\mathcal{C}}(M, X) \to \mathrm{Hom}_{\mathcal{C}}(M, Y) \to \mathrm{Hom}_{\mathcal{C}}(M, Z)$$

is exact. By a) we know the composition is zero. Now suppose given $g \in \mathrm{Hom}_{\mathcal{C}}(M, Y)$ such that $vg = 0$. We apply (TR3) to the triangles $(M, M, 0, \mathrm{id}_M, 0, 0)$ and (X, Y, Z, u, v, w) and the map $g\colon M \to Y$ to conclude there exists $f\colon M \to X$ such that $uf = g$.

A similar proof shows that $\mathrm{Hom}_{\mathcal{C}}(\cdot, M)$ is a contravariant cohomological functor.

c) In the situation of (TR3), suppose f, g are isomorphisms. Apply the cohomological functor $\mathrm{Hom}_{\mathcal{C}}(Z', \cdot)$ to obtain an exact commutative diagram where $f_*, g_*, T(f)_*, T(g)_*$ are isomorphisms of abelian groups. By the five-lemma, h_* is an isomorphism. Hence there exists $\varphi \in \mathrm{Hom}_{\mathcal{C}}(Z', Z)$ such that $h\varphi = \mathrm{id}_Z$. Using $\mathrm{Hom}_{\mathcal{C}}(\cdot, Z)$ similarly gives $\psi \in \mathrm{Hom}_{\mathcal{C}}(Z, Z')$ such that $\psi h = \mathrm{id}_{Z'}$. Thus h is an isomorphism. □

2.3 $\mathcal{K}(\mathcal{A})$ is triangulated

Let $\mathcal{A}$ be an abelian category. A *complex* of objects of $\mathcal{A}$ is a collection $X^\bullet = (X^n)_{n\in\mathbb{Z}}$ of objects of $\mathcal{A}$ together with maps $d^n\colon X^n \to X^{n+1}$ such that $d^{n+1}d^n = 0$ for all $n \in \mathbb{Z}$.

A morphism f of complexes $X^\bullet \to Y^\bullet$ is a collection of maps $f^n\colon X^n \to Y^n$ which commute with differentials:

$$f^{n+1}d_X^n = d_Y^n f^n$$

for all n.

Two maps $f, g\colon X^\bullet \to Y^\bullet$ are *homotopic* if there exists a collection of maps $k = (k^n)$, $k^n\colon X^n \to X^{n-1}$ (not necessarily commuting with d) such that

$$f^n - g^n = d_Y^{n-1}k^n + k^{n+1}d_X^n$$

for all n. Homotopy is an equivalence relation, and compositions of homotopic maps are homotopic.

We define $\mathcal{K}(\mathcal{A})$ to be the category whose objects are complexes of objects of $\mathcal{A}$, and whose morphisms are homotopy equivalence classes of morphisms of complexes.

A complex $X^\bullet$ is *bounded below* if $X^n = 0$ for $n \ll 0$. We denote by $\mathcal{K}^+(\mathcal{A})$ the full subcategory of $\mathcal{K}(\mathcal{A})$ of complexes bounded below. Similarly we define $\mathcal{K}^-(\mathcal{A})$ (bounded above) and $\mathcal{K}^b(\mathcal{A})$ (bounded on both sides).

We now define a triangulated structure on $\mathcal{K}(\mathcal{A})$ (resp. $\mathcal{K}^+(\mathcal{A})$, etc.). Let T be the left shift functor with sign change:

$$T(X^\bullet)^p = X^{p+1}, \qquad d_{T(X)} = -d_X.$$

We often write $X^\bullet[1]$ instead of $T(X^\bullet)$, and $X^\bullet[n]$ instead of $T^n(X^\bullet)$.

If $u\colon X^\bullet \to Y^\bullet$ is any morphism, consider its *mapping cone* $Z^\bullet$, defined as the complex

$$Z^\bullet = T(X^\bullet) \oplus Y^\bullet$$

with differential

$$d_Z = \begin{pmatrix} T(d_X) & T(u) \\ 0 & d_Y \end{pmatrix}.$$

There are natural morphisms

$$v\colon Y^\bullet \to Z^\bullet, \qquad w\colon Z^\bullet \to T(X^\bullet).$$

We define a triangle in $\mathcal{K}(\mathcal{A})$ to be any sextuple isomorphic to one obtained from a morphism u by this construction.

All axioms (TR1)–(TR4) are easily verified, once one notes that the mapping cone of $\mathrm{id}_{X^\bullet}$ is homotopic to 0. Indeed, the matrix

$$k = \begin{pmatrix} 0 & 0 \\ \mathrm{id}_X & 0 \end{pmatrix}$$

gives a contraction homotopy.

Definition 3. *We define H to be the functor from $\mathcal{K}(\mathcal{A})$ to $\mathcal{A}$ which sends a complex $X^\bullet$ to its 0^{th} cohomology:*

$$H(X^\bullet) = \ker d^0 \big/ \operatorname{im} d^{-1}.$$

This is indeed a functor, since homotopic maps induce the same map on cohomology. We write $H^i = H \circ T^i$ for any $i \in \mathbb{Z}$.

Observe that H is a cohomological functor from $\mathcal{K}(\mathcal{A})$ to $\mathcal{A}$. Indeed, it suffices to check the long exact sequence for triangles coming from mapping cones, which is straightforward.

2.4 Localization of categories

Definition 4. *Let $\mathcal{C}$ be a category. A collection S of morphisms of $\mathcal{C}$ is called a* multiplicative system *if it satisfies the following axioms (FR1)–(FR3):*

(FR1) *If $f, g \in S$ and the composite fg exists, then $fg \in S$. For any $X \in \operatorname{Ob}\mathcal{C}$, $\mathrm{id}_X \in S$.*

(FR2) *Any diagram*

$$X \xrightarrow{u} Y \xleftarrow{s} Z, \quad s \in S$$

can be completed to a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{v} & Z \\ {\scriptstyle t}\big\downarrow & & \big\downarrow{\scriptstyle s} \\ X & \xrightarrow{u} & Y \end{array}$$

with $t \in S$. The analogous statement holds with all arrows reversed.

(FR3) *Let $f, g \colon X \to Y$ be morphisms in $\mathcal{C}$. The following are equivalent:*

- *(i) There exists $s \colon Y \to Y'$ in S such that $sf = sg$;*
- *(ii) There exists $t \colon X' \to X$ in S such that $ft = gt$.*

Definition 5. *Let $\mathcal{C}$ be a category and S a multiplicative system of morphisms in $\mathcal{C}$. The* localization of $\mathcal{C}$ with respect to S *is a category $\mathcal{C}_S$ together with a functor $Q \colon \mathcal{C} \to \mathcal{C}_S$ such that:*

- *(a) $Q(s)$ is an isomorphism for every $s \in S$;*
- *(b) Any functor $F \colon \mathcal{C} \to \mathcal{D}$ sending each $s \in S$ to an isomorphism factors uniquely through Q.*

Proposition 2. *Let $\mathcal{C}$ be a category and S a multiplicative system in $\mathcal{C}$. The localization $\mathcal{C}_S$ may be constructed as follows: $\operatorname{Ob}\mathcal{C}_S = \operatorname{Ob}\mathcal{C}$, and for all $X, Y \in \operatorname{Ob}\mathcal{C}$,*

$$\operatorname{Hom}_{\mathcal{C}_S}(X, Y) = \varinjlim_{I_X} \operatorname{Hom}_{\mathcal{C}}(X', Y),$$

where I_X is the category whose objects are morphisms $s \colon X' \to X$ in S, and morphisms are commutative triangles over X.

Furthermore, if $\mathcal{C}$ is additive, then $\mathcal{C}_S$ is also additive.

Proof. Using (FR1), (FR2), (FR3), the index category I_X satisfies the inductive limit axioms (L1, L2, L3) of [GT, Ch. I, §1], so the direct limit is well-behaved.

A morphism $X \to Y$ in $\mathcal{C}_S$ is represented by a right fraction $X \xleftarrow{s} X' \xrightarrow{f} Y$ with $s \in S$. Two such fractions define the same morphism iff there exist $u\colon X''' \to X'$, $g\colon X''' \to X''$ in S making the obvious diagram commute and equalizing the representatives.

Composition of fractions is defined via (FR2): given fractions $X \xleftarrow{s} X' \xrightarrow{a} Y$ and $Y \xleftarrow{t} Y' \xrightarrow{b} Z$, complete the diagram to get a common lift and form the composite morphism $X \to Z$. The result is independent of representative and lift.

The localization functor $Q\colon \mathcal{C} \to \mathcal{C}_S$ satisfies the universal property, and additivity of $\mathcal{C}$ passes to $\mathcal{C}_S$ since the limit preserves abelian group structure on Hom-sets. □

Definition 6. *Let $\mathcal{C}$ be a triangulated category and S a multiplicative system in $\mathcal{C}$. We say S is* compatible with the triangulation *if:*

(FR4) $s \in S \implies T(s) \in S$, where T is the translation functor;

(FR5) Axiom (TR3) holds with the extra condition: if $f, g \in S$ are the given morphisms of triangles, then one may choose the filler $h \in S$.

Proposition 3. *If $\mathcal{C}$ is triangulated and S a triangulation-compatible multiplicative system, then $\mathcal{C}_S$ admits a* unique *triangulated structure such that $Q\colon \mathcal{C} \to \mathcal{C}_S$ is a triangle functor, and Q satisfies the universal property for triangle functors into other triangulated categories.*

One may also compute morphism sets as

$$\mathrm{Hom}_{\mathcal{C}_S}(X, Y) = \varinjlim_{J_Y} \mathrm{Hom}_{\mathcal{C}}(X, Y'),$$

where J_Y has objects $s\colon Y \to Y' \in S$ and morphisms commutative diagrams under Y.

Proposition 4. *Let $\mathcal{C}$ be a category, S a multiplicative system in $\mathcal{C}$, and $\mathcal{D} \subseteq \mathcal{C}$ a full subcategory such that $S \cap \mathcal{D}$ is a multiplicative system in $\mathcal{D}$. Assume one of the following:*

(i) For every $s\colon X' \to X$ in S with $X \in \mathrm{Ob}\,\mathcal{D}$, there exists $f\colon X'' \to X'$ with $X'' \in \mathrm{Ob}\,\mathcal{D}$ and $sf \in S$;

(ii) The same with all arrows reversed.

Then the natural functor

$$\mathcal{D}_{S\cap\mathcal{D}} \to \mathcal{C}_S$$

is fully faithful, so $\mathcal{D}_{S\cap\mathcal{D}}$ may be identified with a full subcategory of $\mathcal{C}_S$.

Proposition 5. *Let $\mathcal{C}$ be a category, S a multiplicative system, $Q\colon \mathcal{C} \to \mathcal{C}_S$ the localization functor. Let $\mathcal{D}$ be any category, and $F, G\colon \mathcal{C}_S \to \mathcal{D}$ functors. Then the natural restriction map*

$$\alpha\colon \operatorname{Hom}(F, G) \to \operatorname{Hom}(FQ, GQ)$$

between functor morphism sets is bijective.

Proof. A functor morphism $F \to G$ assigns to each object $X \in \mathcal{C}_S$ a morphism $F(X) \to G(X)$ making all morphism squares commute.

Injectivity of α is clear since $\operatorname{Ob}\mathcal{C}_S = \operatorname{Ob}\mathcal{C}$. For surjectivity, take a morphism $FQ \to GQ$. Since every morphism in $\mathcal{C}_S$ is a fraction built from morphisms of $\mathcal{C}$ and elements of S, and $F(s), G(s)$ are isomorphisms for $s \in S$, the given functor morphism on FQ, GQ extends uniquely to a functor morphism on F, G over $\mathcal{C}_S$. $\square$

2.5 Quasi-isomorphisms and the derived category

Let $\mathcal{A}$ be an abelian category, and let $\mathrm{K}(\mathcal{A})$ be the triangulated category of complexes up to homotopy introduced in §2. A morphism $f\colon X^\bullet \to Y^\bullet$ in $\mathrm{K}(\mathcal{A})$ is called a *quasi-isomorphism* if it induces an isomorphism on all cohomology groups. Denote by Qis the collection of all quasi-isomorphisms in $\mathrm{K}(\mathcal{A})$.

Proposition 6. *Let $\mathcal{C}$ be a triangulated category, $\mathcal{A}$ an abelian category, and $H\colon \mathcal{C} \to \mathcal{A}$ a cohomological functor. Let S be the class of morphisms $s \in \mathcal{C}$ such that $H(T^i(s))$ is an isomorphism for all $i \in \mathbb{Z}$. Then S is a multiplicative system in $\mathcal{C}$ and is compatible with the triangulation.*

Proof. We verify axioms (FR1)–(FR5). Axioms (FR1) and (FR4) are trivial. Axiom (FR5) follows from the long exact sequence of a cohomological functor and the five-lemma.

To prove (FR2), consider a diagram

$$X \xrightarrow{u} Y \xleftarrow{s} Z, \qquad s \in S.$$

Complete s to a triangle (Z, Y, N, s, f, g). Complete fu to a triangle (W, X, N, t, fu, h). Since (u, id_N) maps two sides of the second triangle to the first, there exists a morphism $v\colon W \to Z$ giving a morphism of triangles, satisfying $sv = ut$.

It remains to show $t \in S$. Since $s \in S$, the long exact sequence of the first triangle implies $H(T^i(N)) = 0$ for all $i \in \mathbb{Z}$. Substituting into the long exact sequence of the second triangle yields that $H(T^i(t))$ is an isomorphism for all $i \in \mathbb{Z}$, so $t \in S$. The dual statement of (FR2) is proved similarly.

For (FR3), consider $f, g\colon X \to Y$. It suffices to prove equivalence of:

(i) $\exists\, s\colon Y \to Y' \in S$ with $sf = sg$;

(ii) $\exists\, t\colon X' \to X \in S$ with $ft = gt$.

It is enough to treat the case $g = 0$, so we need: (i) $\exists\, s\colon Y \to Y' \in S,\ sf = 0 \iff$ (ii) $\exists\, t\colon X' \to X \in S,\ ft = 0$.

Assume (i). By (TR1),(TR2), form a triangle (Z, Y, Y', v, s, u). Since $sf = 0$, Proposition 1.1(b) gives $g\colon X \to Z$ with $f = vg$. Form a triangle (X', X, Z, t, g, w). Again by Proposition 1.1(b), $ft = 0$. Since $s \in S$, $H(T^i(Z)) = 0$ for all i, which implies $t \in S$. The converse (ii)$\Rightarrow$(i) is symmetric. $\square$

Corollary 1. *The set* Qis *is a multiplicative system in* $\mathrm{K}(\mathcal{A})$*, compatible with the triangulation.*

Definition 7. *The* derived category *of $\mathcal{A}$ is*

$$\mathrm{D}(\mathcal{A}) := \mathrm{K}(\mathcal{A})_{\mathrm{Qis}}.$$

Similarly define

$$\mathrm{D}^+(\mathcal{A}) = \mathrm{K}^+(\mathcal{A})_{\mathrm{Qis}}, \quad \mathrm{D}^-(\mathcal{A}) = \mathrm{K}^-(\mathcal{A})_{\mathrm{Qis}}, \quad \mathrm{D}^b(\mathcal{A}) = \mathrm{K}^b(\mathcal{A})_{\mathrm{Qis}}.$$

By Proposition 1.3.3, these are full subcategories of $\mathrm{D}(\mathcal{A})$*, and*

$$\mathrm{D}^+(\mathcal{A}) \cap \mathrm{D}^-(\mathcal{A}) = \mathrm{D}^b(\mathcal{A}).$$

Definition 8. *Let $\mathcal{A}$ be abelian, and $\mathcal{A}' \subset \mathcal{A}$ a* thick *full abelian subcategory (closed under extensions in $\mathcal{A}$). Define* $\mathrm{K}_{\mathcal{A}'}(\mathcal{A})$ *to be the full subcategory of* $\mathrm{K}(\mathcal{A})$ *consisting of complexes $X^\bullet$ whose cohomology objects $H^i(X^\bullet)$ lie in $\mathcal{A}'$. Since $\mathcal{A}'$ is thick,* $\mathrm{K}_{\mathcal{A}'}(\mathcal{A})$ *is a triangulated subcategory of* $\mathrm{K}(\mathcal{A})$.

Define

$$\mathrm{D}_{\mathcal{A}'}(\mathcal{A}) := \mathrm{K}_{\mathcal{A}'}(\mathcal{A})_{\mathrm{Qis}}.$$

By Proposition 1.3.3, $\mathrm{D}_{\mathcal{A}'}(\mathcal{A})$ *is the full subcategory of* $\mathrm{D}(\mathcal{A})$ *of complexes with all cohomology in $\mathcal{A}'$. One defines similarly* $\mathrm{K}^+_{\mathcal{A}'}(\mathcal{A}), \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$*, etc., for bounded-below complexes.*

Definition 9. *A morphism $f\colon X^\bullet \to Y^\bullet$ of complexes induces the zero morphism in* $\mathrm{D}(\mathcal{A})$ *if and only if:*

($*$) *There exists $s\colon Y^\bullet \to Y'^\bullet \in \mathrm{Qis}$ such that sf is homotopic to zero (equivalently, $\exists\, t\colon X'^\bullet \to X^\bullet \in \mathrm{Qis}$ such that ft is homotopic to zero).*

1. If f is homotopic to zero, then ($*$) holds. The converse is false. Take the complex
$$X^\bullet:\ 0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0,$$
and $f = \mathrm{id}_{X^\bullet}$. Let $g\colon X^\bullet \to 0$ be the zero map; then $g \in \mathrm{Qis}$ and $gf = 0$, but $\mathrm{id}_{X^\bullet}$ is not homotopic to zero.

2. If f is homotopic to zero, it induces the zero map on cohomology; the converse is false. Take
$$X^\bullet, Y^\bullet:\quad 0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to 0.$$
The identity map induces zero on cohomology, but there is no $t \in \mathrm{Qis}$ with $ft \simeq 0$. Any homotopy operator would force $2k(x) = 1$, impossible in $\mathbb{Z}$.

For morphisms $f, g\colon X^\bullet \to Y^\bullet$ of complexes, the implications below are all strict:
$$f = g \implies f \simeq g \implies f = g \text{ in } \mathrm{D}(\mathcal{A}) \implies H^i(f) = H^i(g)\ \forall i.$$

Proposition 7. *Let $\mathcal{A}$ be identified with the full subcategory of $\mathrm{D}(\mathcal{A})$ consisting of complexes concentrated in degree zero. Then the natural embedding $\mathcal{A} \hookrightarrow \mathrm{D}(\mathcal{A})$ is fully faithful, i.e., for any $X, Y \in \mathrm{Ob}\,\mathcal{A}$,*
$$\mathrm{Hom}_{\mathcal{A}}(X, Y) \cong \mathrm{Hom}_{\mathrm{D}(\mathcal{A})}(X, Y).$$

We now prove three lemmas, which give an alternative description of $\mathrm{D}^+(\mathcal{A})$ when $\mathcal{A}$ has enough injectives.

Lemma 1. *Let $\mathcal{A}$ be abelian, $f\colon Z^\bullet \to I^\bullet$ a morphism of complexes. Assume:*

(1) $Z^\bullet$ is acyclic;

(2) Each I^p is injective in $\mathcal{A}$;

(3) $I^\bullet$ is bounded below.

Then f is homotopic to zero.

Lemma 2. *Let $\mathcal{A}$ be abelian, $s\colon I^\bullet \to Y^\bullet$ a morphism of complexes. Assume:*

(1) s is a quasi-isomorphism;

(2) Each I^p is injective;

(3) $I^\bullet$ is bounded below.

Then s admits a homotopy inverse.

Proof. Form the mapping cone $Z^\bullet = T(I^\bullet) \oplus Y^\bullet$ of s. Since s is a quasi-isomorphism, $Z^\bullet$ is acyclic. The natural map $v\colon Z^\bullet \to T(I^\bullet)$ satisfies the hypotheses of Lemma 1.4.4, hence is homotopic to zero. Let $(k,t)\colon T(I^\bullet) \oplus Y^\bullet \to I^\bullet$ be the corresponding homotopy. Then

$$v = (\mathrm{id}_{I^\bullet}, 0) = (k,t) \circ d_Z + d_I \circ (k,t).$$

Separating components yields

$$\mathrm{id}_I = dk + kd + ts, \qquad dt - td = 0.$$

Thus $t\colon Y^\bullet \to I^\bullet$ is a morphism of complexes, and ts is homotopic to id_I. Hence t is a homotopy inverse of s. □

Lemma 3. *Let $\mathcal{A}$ be an abelian category.*

1) Let $P \subseteq \operatorname{Ob} \mathcal{A}$ satisfy:

(i) Every object of $\mathcal{A}$ admits an injection into an element of P.

Then every bounded-below complex $X^\bullet \in \mathrm{K}^+(\mathcal{A})$ admits a quasi-isomorphism into a bounded-below complex $I^\bullet$ of objects of P.

2) Assume further that P satisfies:

(ii) If $0 \to X \to Y \to Z \to 0$ is exact and $X \in P$, then $Y \in P \iff Z \in P$;

(iii) There exists a positive integer n such that for any exact sequence

$$X^0 \to X^1 \to \cdots \to X^{n-1} \to X^n \to 0,$$

if $X^0, \ldots, X^{n-1} \in P$ then $X^n \in P$.

Then every $X^\bullet \in \mathrm{K}(\mathcal{A})$ admits a quasi-isomorphism into a complex $I^\bullet$ of objects of P.

3) Let $\mathcal{A}'$ be a thick subcategory of $\mathcal{A}$, and suppose $\mathcal{A}'$ has enough $\mathcal{A}$-injectives. Then every $X^\bullet \in \mathrm{K}^+_{\mathcal{A}'}(\mathcal{A})$ admits a quasi-isomorphism into a bounded-below complex $I^\bullet$ of $\mathcal{A}$-injective objects of $\mathcal{A}'$.

Proof of 1). We may assume $X^p = 0$ for $p < 0$. Embed $X^0 \hookrightarrow I^0$ with $I^0 \in P$. Having defined $I^0, I^1, \dots, I^p$, choose $I^{p+1} \in P$ containing

$$I^p \big/ \operatorname{im} I^{p-1} \;\oplus\; X^{p+1},$$

and define differentials $I^p \to I^{p+1}$ and morphisms $X^{p+1} \to I^{p+1}$ canonically. One verifies $X^\bullet \to I^\bullet$ is a quasi-isomorphism, and each $X^p \to I^p$ is injective. □

Proof of 2). Let $X^\bullet$ be any complex. For a fixed integer i_0, by (i), the truncated complex

$$0 \to 0 \to X^{i_0} \to X^{i_0+1} \to \cdots$$

admits a quasi-isomorphism into a complex $I^\bullet$ of objects of P, with each $X^i \to I^i$ injective. Define $X_0^\bullet$ by

$$\cdots \to X^{i_0-2} \to X^{i_0-1} \to I^{i_0} \to I^{i_0+1} \to \cdots .$$

Then $X^\bullet \to X_0^\bullet$ is a quasi-isomorphism, $X_0^i \in P$ for $i \geq i_0$, and each $X^i \to X_0^i$ is injective.

Now suppose we have $X_1^\bullet$ with $X_1^i \in P$ for $i \geq i_1$, and let $i_2 < i_1$. We construct a quasi-isomorphism $X_1^\bullet \to X_2^\bullet$ such that $X_2^i \in P$ for $i \geq i_2$ and $X_1^i = X_2^i$ for $i \geq i_1 + n$ (with n from condition (iii)).

By the first step, choose a quasi-isomorphism $X_1^\bullet \to X'^\bullet$ with $X'^i \in P$ for $i \geq i_2$ and each map injective. Let $Y^i = \operatorname{coker}(X_1^i \to X'^i)$. Then $Y^\bullet$ is acyclic, and $Y^i \in P$ for $i \geq i_1$ by (ii). By (iii), the boundaries $B^i(Y^\bullet) \in P$ for $i \geq i_1 + n$.

Define $X_2^\bullet$ piecewise:

$$X_2^i = \begin{cases} X'^i & i < i_1 + n, \\ B^i(X'^\bullet) \oplus \ker(X_1^{i-1} \to X_1^i) & i = i_1 + n, \\ X_1^i & i > i_1 + n. \end{cases}$$

From the exact sequence

$$0 \to X_1^i \to B^i(X'^\bullet) \oplus \ker(X_1^{i-1} \to X_1^i) \to B^i(X'^\bullet) \to 0,$$

condition (ii) implies the middle term lies in P for $i \geq i_1 + n$. One checks $X_1^\bullet \to X_2^\bullet$ is a quasi-isomorphism.

Now for any $X^\bullet \in \mathrm{K}(\mathcal{A})$, pick a decreasing sequence $i_0 > i_1 > i_2 > \cdots \to -\infty$. Construct $X_0^\bullet, X_1^\bullet, X_2^\bullet, \dots$ as above, giving quasi-isomorphisms

$$X^\bullet \to X_0^\bullet \to X_1^\bullet \to X_2^\bullet \to \cdots .$$

For each fixed degree i, the sequence $X^i \to X_0^i \to X_1^i \to \cdots$ is eventually constant and lies in P. The direct limit $I^\bullet = \varinjlim X_r^\bullet$ is the desired complex of objects of P. □

Proof of 3). Assume $X^i = 0$ for $i < 0$. Embed $H^0(X^\bullet) \hookrightarrow I^0$, where I^0 is an $\mathcal{A}$-injective object of $\mathcal{A}'$; this is possible since $H^0(X^\bullet) \in \mathcal{A}'$ and $\mathcal{A}'$ has enough $\mathcal{A}$-injectives. Extend to a morphism $f^0 \colon X^0 \to I^0$.

Inductively having defined $I^0, \ldots, I^p$ and $f^i \colon X^i \to I^i$ for $0 \le i \le p$, choose an $\mathcal{A}$-injective $I^{p+1} \in \mathcal{A}'$ containing

$$I^p \big/ \operatorname{im} I^{p-1} \;\oplus\; Z^{p+1}(X^\bullet).$$

Since $\mathcal{A}'$ is thick, this object belongs to $\mathcal{A}'$. Extend the natural map $Z^{p+1}(X^\bullet) \to I^{p+1}$ to $f^{p+1} \colon X^{p+1} \to I^{p+1}$. The resulting morphism $f \colon X^\bullet \to I^\bullet$ is a quasi-isomorphism. □

Proposition 8. *Let $\mathcal{A}$ be abelian, and let $\mathcal{I}$ be the additive subcategory of injective objects of $\mathcal{A}$. The natural functor*

$$\alpha \colon \mathrm{K}^+(\mathcal{I}) \to \mathrm{D}^+(\mathcal{A})$$

is fully faithful. If furthermore $\mathcal{A}$ has enough injectives, then α is an equivalence of categories.

Proof. By Proposition 1.4.2, $\mathrm{Qis} \cap \mathrm{K}^+(\mathcal{I})$ is a multiplicative system in $\mathrm{K}^+(\mathcal{I})$. Lemma 1.4.5 verifies condition (ii) of Proposition 1.3.3 for the inclusion $\mathrm{K}^+(\mathcal{I}) \subseteq \mathrm{K}^+(\mathcal{A})$ and Qis. Thus

$$\mathrm{D}^+(\mathcal{I}) \to \mathrm{D}^+(\mathcal{A})$$

is fully faithful. By Lemma 1.4.5, every quasi-isomorphism in $\mathrm{K}^+(\mathcal{I})$ is an isomorphism, so $\mathrm{K}^+(\mathcal{I}) = \mathrm{D}^+(\mathcal{I})$.

If $\mathcal{A}$ has enough injectives, apply Lemma 1.4.6(1) with $P = \operatorname{Ob}\mathcal{I}$: every object of $\mathrm{D}^+(\mathcal{A})$ is isomorphic to an object of $\mathrm{K}^+(\mathcal{I})$. Hence α is an equivalence. □

Proposition 9. *Let $\mathcal{A}$ be abelian, $\mathcal{A}' \subseteq \mathcal{A}$ a thick abelian subcategory. Assume $\mathcal{A}'$ has enough $\mathcal{A}$-injectives. Then the natural functor*

$$\mathrm{D}^+(\mathcal{A}') \to \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$$

is an equivalence of categories.

Proof. Apply Proposition 1.3.3 to the inclusion $\mathrm{K}^+(\mathcal{A}') \hookrightarrow \mathrm{K}^+(\mathcal{A})$. The set Qis is a multiplicative system in both categories. Any quasi-isomorphism $X^\bullet \to Y^\bullet$ with $X^\bullet \in \mathrm{K}^+(\mathcal{A}')$ has cohomology in $\mathcal{A}'$; by Lemma 1.4.6(3), $Y^\bullet$ admits a quasi-isomorphism into an injective complex $I^\bullet \in \mathrm{K}^+(\mathcal{A}')$. Thus condition (ii) of Proposition 1.3.3 holds, so $\mathrm{D}^+(\mathcal{A}') \to \mathrm{D}^+(\mathcal{A})$ is fully faithful, with image exactly $\mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$. $\square$

Exercise. Formulate the analogous statements for projective objects of $\mathcal{A}$ and the bounded-above derived category $\mathrm{D}^-(\mathcal{A})$.

2.6 Derived functors

We treat only *right derived covariant functors*; the reader may make the obvious modifications for left derived covariant functors, and right/left derived contravariant functors.

Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and let $F\colon \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$ be a triangulated functor (cf. §1). This holds in particular for any additive functor $F\colon \mathcal{A} \to \mathcal{B}$, which extends canonically to $\mathrm{K}(\mathcal{A})$.

In general, F need not send quasi-isomorphisms to quasi-isomorphisms; if it does, then F localizes to induce a functor $\mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$. Exact functors are examples of this behavior.

We are led to ask whether there exists a functor $\mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ approximating F; this motivates the general definition of derived functors below.

Definition 10. *Let $\mathcal{A}$ be abelian, and let $\mathrm{K}^*(\mathcal{A}) \subseteq \mathrm{K}(\mathcal{A})$ be a triangulated subcategory. By Proposition 1.4.2, $\mathrm{K}^*(\mathcal{A}) \cap \mathrm{Qis}$ is a multiplicative system in $\mathrm{K}^*(\mathcal{A})$.*

We say $\mathrm{K}^*(\mathcal{A})$ is a localizing subcategory *of $\mathrm{K}(\mathcal{A})$ if the natural functor*

$$\mathrm{K}^*(\mathcal{A})_{\mathrm{K}^*(\mathcal{A})\cap\mathrm{Qis}} \;\to\; \mathrm{K}(\mathcal{A})_{\mathrm{Qis}} = \mathrm{D}(\mathcal{A})$$

is fully faithful. In this case we write $\mathrm{D}^*(\mathcal{A})$ for the domain category above.

Examples.

1. The intersection of any family of localizing subcategories is localizing.

2. $\mathrm{K}^+(\mathcal{A}), \mathrm{K}^-(\mathcal{A}), \mathrm{K}^b(\mathcal{A})$ are localizing subcategories of $\mathrm{K}(\mathcal{A})$ (cf. §4).

3. If $\mathcal{A}' \subseteq \mathcal{A}$ is a thick subcategory, then $\mathrm{K}_{\mathcal{A}'}(\mathcal{A}), \mathrm{K}^+_{\mathcal{A}'}(\mathcal{A}), \mathrm{K}^-_{\mathcal{A}'}(\mathcal{A}), \mathrm{K}^b_{\mathcal{A}'}(\mathcal{A})$ are localizing subcategories of $\mathrm{K}(\mathcal{A})$ (cf. §4).

4. The complexes of finite injective dimension form a localizing subcategory $\mathrm{K}^+(\mathcal{A})_{\mathrm{fid}} \subseteq \mathrm{K}(\mathcal{A})$ (cf. Corollary 1.7.7).

Definition 11. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $\mathrm{K}^*(\mathcal{A}) \subseteq \mathrm{K}(\mathcal{A})$ a localizing subcategory, and*

$$F\colon \mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$$

a triangulated functor.

Denote by Q the localization functor $\mathrm{K}^*(\mathcal{A}) \to \mathrm{D}^*(\mathcal{A})$ (resp. $\mathrm{K}(\mathcal{B}) \to \mathrm{D}(\mathcal{B})$). The right derived functor *of F is a pair*

$$\mathbf{R}^*F\colon \mathrm{D}^*(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$$

together with a functor morphism

$$\xi\colon Q \circ F \ \to \ \mathbf{R}^*F \circ Q$$

of functors $\mathrm{K}^*(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$, satisfying the following universal property:

For any triangulated functor $G\colon \mathrm{D}^*(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ and any functor morphism

$$\zeta\colon Q \circ F \ \to \ G \circ Q,$$

there exists a unique functor morphism

$$\eta\colon \mathbf{R}^*F \to G$$

such that

$$\zeta = (\eta \circ Q) \circ \xi.$$

Remarks.

1. If $\mathbf{R}^*F$ exists, it is unique up to unique isomorphism of functors.

2. For standard localizing subcategories $\mathrm{K}^+(\mathcal{A}), \mathrm{K}^-(\mathcal{A}), \mathrm{K}_{\mathcal{A}'}(\mathcal{A})$, we write

$$\mathbf{R}^+F, \quad \mathbf{R}^-F, \quad \mathbf{R}_{\mathcal{A}'}F$$

   respectively; when no confusion arises, we write simply $\mathbf{R}F$.

3. We define $R^iF = H^i(\mathbf{R}F)$. If F is left-exact on $\mathcal{A}$ and $\mathcal{A}$ has enough injectives, these coincide with the classical right derived functors of F.

4. Let $\varphi\colon F \to G$ be a morphism of functors $\mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$. If $\mathbf{R}F, \mathbf{R}G$ exist, there is a unique induced morphism

$$\mathbf{R}\varphi\colon \mathbf{R}F \to \mathbf{R}G$$

compatible with the structure morphisms ξ. This follows directly from the universal property.

5. Let $\mathrm{K}^{**}(\mathcal{A}) \subseteq \mathrm{K}^*(\mathcal{A})$ be localizing subcategories, $F\colon\ \mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$ a triangulated functor. If both $\mathbf{R}^*F$ and $\mathbf{R}^{**}(F|_{\mathrm{K}^{**}(\mathcal{A})})$ exist, there is a natural functor morphism

$$\mathbf{R}^{**}\big(F|_{\mathrm{K}^{**}(\mathcal{A})}\big) \ \to\ \mathbf{R}^*F\big|_{\mathrm{D}^{**}(\mathcal{A})}.$$

We do not know if this is an isomorphism in general, but it holds in all our intended applications (cf. Corollary 1.5.3).

Theorem 1 (Existence of Derived Functors)**.** *Let $\mathcal{A}, \mathcal{B}, \mathrm{K}^*(\mathcal{A}), F$ be as above. Suppose there exists a triangulated subcategory $\mathrm{L} \subseteq \mathrm{K}^*(\mathcal{A})$ such that:*

(1) Every object of $\mathrm{K}^*(\mathcal{A})$ admits a quasi-isomorphism into some object of L;

(2) For every acyclic complex $I^\bullet \in \operatorname{Ob}\mathrm{L}$ (i.e. $H^i(I^\bullet) = 0$ for all i), the complex $F(I^\bullet)$ is also acyclic.

Then the right derived functor $(\mathbf{R}^*F, \xi)$ exists. Furthermore, for any $I^\bullet \in \operatorname{Ob}\mathrm{L}$, the structure morphism

$$\xi(I^\bullet)\colon Q \circ F(I^\bullet) \to \mathbf{R}^*F \circ Q(I^\bullet)$$

is an isomorphism in $\mathrm{D}(\mathcal{B})$.

Proof. First, the restriction $F|_{\mathrm{L}}$ sends quasi-isomorphisms to quasi-isomorphisms. Indeed, let $s\colon I_1^\bullet \to I_2^\bullet$ be a quasi-isomorphism in L; complete s to a triangle in $\mathrm{K}^*(\mathcal{A})$. The third vertex $J^\bullet$ of the triangle is acyclic, so by hypothesis (2), $F(J^\bullet)$ is acyclic. Hence $F(s)$ is a quasi-isomorphism.

Thus F descends to a functor

$$\overline{F}\colon\ \mathrm{L}_{\mathrm{Qis}} \to \mathrm{D}(\mathcal{B})$$

satisfying $\overline{F} \circ Q = Q \circ F$ on L (with Q denoting localization functors as usual).

Second, the pair $(\mathrm{L}, \mathrm{K}^*(\mathcal{A}), \mathrm{Qis})$ satisfies condition (ii) of Proposition 1.3.3, so the natural functor

$$T\colon\ \mathrm{L}_{\mathrm{Qis}} \to \mathrm{D}^*(\mathcal{A})$$

is an equivalence of categories. Let $U\colon \mathrm{D}^*(\mathcal{A}) \to \mathrm{L}_{\mathrm{Qis}}$ be a quasi-inverse of T, with functorial isomorphisms

$$\alpha\colon \mathrm{id}_{\mathrm{L}_{\mathrm{Qis}}} \to U \circ T, \qquad \beta\colon \mathrm{id}_{\mathrm{D}^*(\mathcal{A})} \to T \circ U.$$

Define

$$\mathbf{R}^* F := \overline{F} \circ U.$$

We now define the functor morphism

$$\xi\colon Q \circ F \to \mathbf{R}^* F \circ Q = \overline{F} \circ U \circ Q$$

as follows. Given $X^\bullet \in \operatorname{Ob} \mathrm{K}^*(\mathcal{A})$, choose $I^\bullet \in \operatorname{Ob} \mathrm{L}$ such that $Q(I^\bullet) \cong U \circ Q(X^\bullet)$ in $\mathrm{D}^*(\mathcal{A})$.

We have the canonical isomorphism

$$\beta\big(Q(X^\bullet)\big)\colon Q(X^\bullet) \xrightarrow{\sim} T \circ U\big(Q(X^\bullet)\big) = T\big(Q(I^\bullet)\big).$$

This isomorphism may be represented by a diagram in $\mathrm{K}^*(\mathcal{A})$

$$X^\bullet \xleftarrow{s} Y^\bullet \xrightarrow{t} I^\bullet$$

with $s, t \in \mathrm{Qis}$ and $Y^\bullet \in \operatorname{Ob} \mathrm{K}^*(\mathcal{A})$. By hypothesis (1), we may assume $Y^\bullet \in \operatorname{Ob} \mathrm{L}$. Applying the triangulated functor F, we obtain a diagram in $\mathrm{K}(\mathcal{B})$:

$$F(X^\bullet) \xleftarrow{F(s)} F(Y^\bullet) \xrightarrow{F(t)} F(I^\bullet).$$

As observed previously, $F(s), F(t)$ are quasi-isomorphisms.

This induces a morphism in $\mathrm{D}(\mathcal{B})$:

$$\xi(X^\bullet)\colon Q \circ F(X^\bullet) \to Q \circ F(I^\bullet) = \overline{F} \circ Q(I^\bullet) = \overline{F} \circ U \circ Q(X^\bullet) = \mathbf{R}^* F \circ Q(X^\bullet).$$

One verifies that $\xi(X^\bullet)$ is independent of the choice of diagram above, that ξ defines a functor morphism $Q \circ F \to \mathbf{R}^* F \circ Q$, and that the pair $(\mathbf{R}^* F, \xi)$ satisfies the universal property of the right derived functor.

If $X^\bullet \in \operatorname{Ob} \mathrm{L}$, then $F(t)$ is also a quasi-isomorphism in the construction above, so $\xi(X^\bullet)$ is an isomorphism in $\mathrm{D}(\mathcal{B})$, as required. □

Proposition 10. *Let* $\mathcal{A}, \mathcal{B}, \mathrm{K}^*(\mathcal{A}), F$ *be as above, and let* $\mathrm{K}^{**}(\mathcal{A}) \subseteq \mathrm{K}^*(\mathcal{A})$ *be another localizing subcategory of* $\mathrm{K}(\mathcal{A})$. *Suppose there exists a triangulated subcategory* $\mathrm{L} \subseteq \mathrm{K}^*(\mathcal{A})$ *satisfying hypotheses (1),(2) of Theorem 1.5.1, and such that* $\mathrm{L} \cap \mathrm{K}^{**}(\mathcal{A})$ *satisfies hypothesis (1) for* $\mathrm{K}^{**}(\mathcal{A})$. *If* $\mathbf{R}^* F$ *and* $\mathbf{R}^{**}\big(F|_{\mathrm{K}^{**}(\mathcal{A})}\big)$ *both exist, then the natural morphism*

$$\mathbf{R}^{**}\big(F|_{\mathrm{K}^{**}(\mathcal{A})}\big) \ \to\ \mathbf{R}^* F\big|_{\mathrm{D}^{**}(\mathcal{A})}$$

is an isomorphism.

Proof. Existence follows immediately from Theorem 1.5.1. For the isomorphism, any object of $\mathrm{D}^{**}(\mathcal{A})$ is isomorphic to $Q(I^\bullet)$ for some $I^\bullet \in \mathrm{L} \cap \mathrm{K}^{**}(\mathcal{A})$. The claim then follows from the final assertion of Theorem 1.5.1. □

Corollary 2. *α) Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F\colon \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$ a triangulated functor (e.g., induced by an additive functor $F_0\colon \mathcal{A} \to \mathcal{B}$). If $\mathcal{A}$ has enough injectives, then $\mathbf{R}^+F$ exists.*

β) Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F\colon \mathcal{A} \to \mathcal{B}$ additive. Suppose there exists $P \subseteq \mathrm{Ob}\,\mathcal{A}$ satisfying (i),(ii) of Lemma 1.4.6, and:

(iv) F sends short exact sequences in $\mathcal{A}$ to short exact sequences in $\mathcal{B}$.

Then $\mathbf{R}^+F$ exists (we also denote by F its canonical extension $\mathrm{K}^+(\mathcal{A}) \to \mathrm{K}^+(\mathcal{B})$).

γ) Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F\colon \mathcal{A} \to \mathcal{B}$ additive. Assume:

(a) The hypotheses of β hold;

(b) F has finite cohomological dimension: there exists $n > 0$ such that $R^iF(Y) = 0$ for all $Y \in \mathrm{Ob}\,\mathcal{A}$ and all $i > n$.

Then $\mathbf{R}F$ exists on all of $\mathrm{D}(\mathcal{A})$, and its restriction to $\mathrm{D}^+(\mathcal{A})$ coincides with $\mathbf{R}^+F$.

Remark. Case α is a special case of β: if $\mathcal{A}$ has enough injectives, the class of injective objects satisfies (i),(ii),(iv) for any additive F.

Proof. α) Let $\mathrm{L} \subseteq \mathrm{K}^+(\mathcal{A})$ be the triangulated subcategory of bounded-below injective complexes. By Lemma 1.4.6(1), every object of $\mathrm{K}^+(\mathcal{A})$ admits a quasi-isomorphism into an object of L. By Lemma 1.4.5, every quasi-isomorphism in L is an isomorphism, so acyclic objects of L are zero in $\mathrm{K}(\mathcal{A})$. Thus condition (2) of Theorem 1.5.1 holds for any triangulated F, so $\mathbf{R}^+F$ exists.

β) Take $\mathrm{L} \subseteq \mathrm{K}^+(\mathcal{A})$ to be complexes of objects of P; L is triangulated by stability of P under extensions. Condition (1) holds by Lemma 1.4.6(1). For (2), let $Z^\bullet \in \mathrm{L}$ be acyclic; using (ii) repeatedly, all kernels $\ker d_Z^p$ lie in P. By (iv), F preserves exactness, so $F(Z^\bullet)$ is acyclic. Theorem 1.5.1 applies, hence $\mathbf{R}^+F$ exists.

γ) Let P' be the class of F-acyclic objects of $\mathcal{A}$ ($R^iF(X) = 0, \forall i > 0$). Then P' satisfies (i),(ii),(iii) of Lemma 1.4.6. Let $\mathrm{L} \subseteq \mathrm{K}(\mathcal{A})$ be complexes of objects of P'. By Lemma 1.4.6 and the argument for β, the hypotheses of Theorem 1.5.1 are satisfied, so $\mathbf{R}F$ exists.

By Proposition 1.5.2, the restriction of $\mathbf{R}F$ to $\mathrm{D}^+(\mathcal{A})$ is isomorphic to $\mathbf{R}^+F$. □

Proposition 11. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories,* $\mathrm{K}^*(\mathcal{A}) \subseteq \mathrm{K}(\mathcal{A})$, $\mathrm{K}^\dagger(\mathcal{B}) \subseteq \mathrm{K}(\mathcal{B})$ *localizing subcategories, and let*

$$F\colon\ \mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B}), \qquad G\colon\ \mathrm{K}^\dagger(\mathcal{B}) \to \mathrm{K}(\mathcal{C})$$

be triangulated functors.

a) *Assume* $F(\mathrm{K}^*(\mathcal{A})) \subseteq \mathrm{K}^\dagger(\mathcal{B})$*, that* $\mathbf{R}^*F, \mathbf{R}^\dagger G, \mathbf{R}^*(G\circ F)$ *exist, and* $\mathbf{R}^*F(\mathrm{D}^*(\mathcal{A})) \subseteq \mathrm{D}^\dagger(\mathcal{B})$*. Then there exists a unique functor morphism*

$$\zeta\colon \mathbf{R}^*(G \circ F) \to \mathbf{R}^\dagger G \circ \mathbf{R}^* F$$

making the diagram of structure maps ξ commutative.

b) *Suppose there exist triangulated subcategories* $\mathrm{L} \subseteq \mathrm{K}^*(\mathcal{A})$, $\mathrm{M} \subseteq \mathrm{K}^\dagger(\mathcal{B})$ *satisfying Theorem 1.5.1's hypotheses for F, G respectively, and* $F(\mathrm{L}) \subseteq \mathrm{M}$*. Then the hypotheses of (a) hold, and the canonical morphism ζ is an isomorphism.*

Proof. Straightforward from universal properties. □

Remarks.

1. For three composable functors F, G, H, the morphisms ζ yield a commutative diagram of derived functor compositions whenever all derived functors exist.

2. This proposition illustrates the power of derived categories: classical composite functor spectral sequences are replaced by direct functor composition in the derived category. The spectral sequence may be recovered by taking cohomology of the derived functor morphism.

Proposition 12. *Let $\mathcal{A}' \subseteq \mathcal{A}$ be a thick abelian subcategory, $\mathcal{B}$ another abelian category,* $F\colon\ \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}^+(\mathcal{B})$ *a triangulated functor. Suppose* $\mathbf{R}^+F$ *and* $\mathbf{R}^+\big(F|_{\mathcal{A}'}\big)$ *both exist. There is a natural functor morphism*

$$\zeta\colon \mathbf{R}^+\big(F|_{\mathcal{A}'}\big) \to \varphi \circ \mathbf{R}^+F,$$

where $\varphi\colon\ \mathrm{D}^+(\mathcal{A}') \to \mathrm{D}^+(\mathcal{A})$ *is the natural embedding. If additionally $\mathcal{A}'$ has enough $\mathcal{A}$-injectives and $\mathcal{A}$ has enough injectives, then ζ is an isomorphism.*

Proof. Existence of ζ follows from the definition of derived functors. If $\mathcal{A}'$ has enough $\mathcal{A}$-injectives, we may compute both derived functors via injective resolutions in $\mathcal{A}'$; by Proposition 1.4.8, φ is fully faithful, hence ζ is an isomorphism. □

2.7 Examples. Ext and R$\mathcal{H}om$

Definition 12. *Let $\mathcal{A}$ be an abelian category, and let $X^\bullet, Y^\bullet \in \mathrm{D}(\mathcal{A})$. Define the* hyperext groups *by*

$$\mathrm{Ext}^i\left(X^\bullet, Y^\bullet\right) = \mathrm{Hom}_{\mathrm{D}(\mathcal{A})}\left(X^\bullet, T^i(Y^\bullet)\right).$$

Remarks.

1. If $X^\bullet, Y^\bullet \in \mathrm{D}^+(\mathcal{A})$, then Ext^i computed in $\mathrm{D}^+(\mathcal{A})$ agrees with the above, since $\mathrm{D}^+(\mathcal{A})$ is a full subcategory of $\mathrm{D}(\mathcal{A})$.

2. This definition gives $\mathrm{Ext}^i(X, Y)$ for any $X, Y \in \mathcal{A}$. We shall see that if $\mathcal{A}$ has enough injectives, this coincides with the classical definition of Ext groups.

Proposition 13. *Let*

$$0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$$

be a short exact sequence of complexes over $\mathcal{A}$, and let $V^\bullet$ be another complex. Then we have long exact sequences

$$\cdots \to \mathrm{Ext}^i(V^\bullet, X^\bullet) \to \mathrm{Ext}^i(V^\bullet, Y^\bullet) \to \mathrm{Ext}^i(V^\bullet, Z^\bullet) \to \mathrm{Ext}^{i+1}(V^\bullet, X^\bullet) \to \cdots$$

and

$$\cdots \to \mathrm{Ext}^i(Z^\bullet, V^\bullet) \to \mathrm{Ext}^i(Y^\bullet, V^\bullet) \to \mathrm{Ext}^i(X^\bullet, V^\bullet) \to \mathrm{Ext}^{i+1}(Z^\bullet, V^\bullet) \to \cdots$$

Proof. Complete $X^\bullet \to Y^\bullet$ to a triangle in $\mathrm{K}(\mathcal{A})$, with third vertex $W^\bullet$:

$$X^\bullet \to Y^\bullet \to W^\bullet \to T(X^\bullet).$$

There exists a morphism $g\colon W^\bullet \to Z^\bullet$. Using the long exact cohomology sequence and the five-lemma, one verifies g is a quasi-isomorphism, hence an isomorphism in $\mathrm{D}(\mathcal{A})$. Replacing $Z^\bullet$ by $W^\bullet$, the result follows from properties of triangles. □

Remark. Any short exact sequence of complexes

$$0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$$

determines a morphism $Z^\bullet \to T(X^\bullet)$ in $\mathrm{D}(\mathcal{A})$, making $X^\bullet, Y^\bullet, Z^\bullet$ into a triangle.

We now define a complex whose cohomology yields the Ext groups.

Definition 13. *For complexes $X^\bullet, Y^\bullet$ over $\mathcal{A}$, define the complex $\mathcal{H}om^\bullet(X^\bullet, Y^\bullet)$ by*

$$\mathcal{H}om^n\big(X^\bullet, Y^\bullet\big) = \prod_{p\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{A}}\left(X^p, Y^{p+n}\right),$$

with differential

$$d^n = \prod_{p\in\mathbb{Z}} \left(d_X^{p-1} + (-1)^{n+1} d_Y^{p+n}\right).$$

The n-cycles of $\mathcal{H}om^\bullet(X^\bullet, Y^\bullet)$ correspond bijectively to morphisms $X^\bullet \to T^n(Y^\bullet)$ of complexes. The n-boundaries correspond to morphisms homotopic to zero. Thus

$$H^n\big(\mathcal{H}om^\bullet(X^\bullet, Y^\bullet)\big) \cong \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}\left(X^\bullet, T^n(Y^\bullet)\right).$$

Clearly $\mathcal{H}om^\bullet$ is a bifunctor

$$\mathcal{H}om^\bullet\colon\ \mathrm{K}(\mathcal{A})^\circ \times \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathbf{Ab}).$$

If $\mathcal{A}$ has enough injectives, we may form its derived functors.

Lemma 4. *Let $X^\bullet \in \mathrm{K}(\mathcal{A})$, $Y^\bullet \in \mathrm{K}^+(\mathcal{A})$ be a bounded-below complex of injectives. Assume either*

(a) $Y^\bullet$ is acyclic, or

(b) $X^\bullet$ is acyclic.

Then $\mathcal{H}om^\bullet(X^\bullet, Y^\bullet)$ is acyclic.

Proof. It suffices to show any morphism $X^\bullet \to Y^\bullet$ is homotopic to zero. Case (a): An acyclic bounded-below injective complex is split exact; one constructs a homotopy directly. Case (b): Follows from Lemma 1.4.4. $\square$

Suppose $\mathcal{A}$ has enough injectives, and let $\mathrm{L} \subseteq \mathrm{K}^+(\mathcal{A})$ be the triangulated subcategory of bounded-below injective complexes. By Lemma 1.6.2(a), for fixed $X^\bullet \in \mathrm{K}(\mathcal{A})$, the functor

$$\mathcal{H}om^\bullet(X^\bullet, \cdot\,)\colon\ \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}(\mathbf{Ab})$$

satisfies the hypotheses of Theorem 1.5.1, hence admits a right derived functor. This is functorial in $X^\bullet$, giving a biderived functor

$$\mathbf{R}_{\mathrm{II}}\mathcal{H}om^\bullet\colon\ \mathrm{K}(\mathcal{A})^\circ \times \mathrm{D}^+(\mathcal{A}) \to \mathrm{D}(\mathbf{Ab}).$$

By Lemma 1.6.2(b), this functor sends acyclic complexes to acyclic complexes in the first variable, hence descends to

$$\mathbf{R}\mathcal{H}om^\bullet\colon \mathrm{D}(\mathcal{A})^\circ \times \mathrm{D}^+(\mathcal{A}) \to \mathrm{D}(\mathbf{Ab}).$$

Dually, if $\mathcal{A}$ has enough projectives, we obtain

$$\mathbf{R}_{\mathrm{I}}\mathcal{H}om^\bullet\colon \mathrm{D}^-(\mathcal{A})^\circ \times \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathbf{Ab}).$$

If $\mathcal{A}$ has both enough injectives and projectives, the two derived functors are canonically isomorphic on $\mathrm{D}^-(\mathcal{A})^\circ \times \mathrm{D}^+(\mathcal{A})$. We therefore write simply $\mathbf{R}\mathcal{H}om^\bullet$.

Lemma 5. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories,*

$$T\colon \mathrm{K}^*(\mathcal{A}) \times \mathrm{K}^\dagger(\mathcal{B}) \to \mathrm{K}(\mathcal{C})$$

a bitriangulated functor. If the iterated derived functors $\mathbf{R}_{\mathrm{I}}\mathbf{R}_{\mathrm{II}}T$ and $\mathbf{R}_{\mathrm{II}}\mathbf{R}_{\mathrm{I}}T$ both exist, they are canonically isomorphic.

Proof. Immediate from the universal property of derived functors. □

Theorem 2 (Yoneda)**.** *Let $\mathcal{A}$ have enough injectives. For all $X^\bullet \in \mathrm{D}(\mathcal{A})$, $Y^\bullet \in \mathrm{D}^+(\mathcal{A})$,*

$$H^i\big(\mathbf{R}\mathcal{H}om^\bullet(X^\bullet, Y^\bullet)\big) = \mathrm{Ext}^i\left(X^\bullet, Y^\bullet\right).$$

Corollary 3. *If $\mathcal{A}$ has enough injectives, the derived-category definition of* $\mathrm{Ext}^i(X, Y)$ *coincides with the classical Ext groups for all $X, Y \in \mathcal{A}$.*

Proof of Corollary 1.6.5. Take an injective resolution $I^\bullet$ of Y. Then

$$\mathrm{Ext}^i(X, Y) = H^i\big(\mathcal{H}om^\bullet(X, I^\bullet)\big),$$

which is the standard definition. □

Proof of Theorem 1.6.4. Choose a quasi-isomorphism $s\colon Y^\bullet \to I^\bullet$ with $I^\bullet$ a bounded-below injective complex. Then

$$\mathrm{Ext}^i(X^\bullet, Y^\bullet) = \mathrm{Ext}^i(X^\bullet, I^\bullet) = \mathrm{Hom}_{\mathrm{D}(\mathcal{A})}\left(X^\bullet, T^i(I^\bullet)\right).$$

By Lemma 1.4.5, any morphism $X^\bullet \to T^i(I^\bullet)$ in $\mathrm{D}(\mathcal{A})$ is represented by an actual morphism of complexes. Therefore

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{D}(\mathcal{A})}\left(X^\bullet, T^i(I^\bullet)\right) &= \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}\left(X^\bullet, T^i(I^\bullet)\right) \\
&= H^i\big(\mathcal{H}om^\bullet(X^\bullet, I^\bullet)\big) \\
&= H^i\big(\mathbf{R}\mathcal{H}om^\bullet(X^\bullet, Y^\bullet)\big).
\end{aligned}$$

□

2.8 Way-out functors

Definition 14. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and let $F\colon \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ be a triangulated functor. We say F is* way-out right *if for every $n_1 \in \mathbb{Z}$, there exists $n_2 \in \mathbb{Z}$ such that for any complex $X^\bullet \in \mathrm{D}(\mathcal{A})$ satisfying $H^i(X^\bullet) = 0$ for all $i < n_2$, we have $H^i(F(X^\bullet)) = 0$ for all $i < n_1$.*

One defines way-out left *and* way-out in both directions *similarly. For a contravariant triangulated functor $F\colon \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$, the definition of way-out right is the same, except reversing the inequality to $i > n_2$.*

Examples.

1. If $F_0\colon \mathcal{A} \to \mathcal{B}$ is additive and satisfies the hypotheses of Corollary 1.5.3(α) or (β), then $\mathbf{R}^+F$ is way-out right. If F_0 satisfies (γ), then $\mathbf{R}F$ is way-out in both directions.

2. For unbounded $X^\bullet \in \mathrm{D}(\mathcal{A})$, the functor $\mathbf{R}\mathcal{H}om(X^\bullet, \cdot\,)$ on $\mathrm{D}^+(\mathcal{A})$ is generally not way-out.

Proposition 14 (Lemma on Way-out Functors)**.** *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $\mathcal{A}' \subseteq \mathcal{A}$ a thick subcategory. Let $F, G\colon \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ be triangulated functors, and $\eta\colon F \to G$ a morphism of functors.*

(i) If $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$, then $\eta(X^\bullet)$ is an isomorphism for all $X^\bullet \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$.

(ii) If $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$, and F, G are way-out right, then $\eta(X^\bullet)$ is an isomorphism for all $X^\bullet \in \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$.

(iii) If $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$, and F, G are way-out in both directions, then $\eta(X^\bullet)$ is an isomorphism for all $X^\bullet \in \mathrm{D}_{\mathcal{A}'}(\mathcal{A})$.

(iv) Let $P \subseteq \operatorname{Ob}\mathcal{A}'$ be such that every object of $\mathcal{A}'$ embeds into an object of P. If $\eta(X)$ is an isomorphism for all $X \in P$, and F, G are way-out right, then $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$.

Definition 15. *Let $X^\bullet$ be a complex over abelian category $\mathcal{A}$, $n \in \mathbb{Z}$. Define truncation functors:*

$$\begin{aligned}
\tau_{>n}(X^\bullet)&\colon\ \cdots \to 0 \to X^{n+1} \to X^{n+2} \to \cdots\,,\\
\tau_{\le n}(X^\bullet)&\colon\ \cdots \to X^{n-1} \to X^n \to 0 \to \cdots\,,\\
\sigma_{>n}(X^\bullet)&\colon\ \cdots \to 0 \to \operatorname{im} d^n \to X^{n+1} \to X^{n+2} \to \cdots\,,\\
\sigma_{\le n}(X^\bullet)&\colon\ \cdots \to X^{n-1} \to \ker d^n \to 0 \to \cdots\,.
\end{aligned}$$

There are natural morphisms inducing short exact sequences of complexes:

$$0 \to \tau_{>n}(X^\bullet) \to X^\bullet \to \tau_{\le n}(X^\bullet) \to 0, \tag{2.1}$$
$$0 \to \sigma_{\le n}(X^\bullet) \to X^\bullet \to \sigma_{>n}(X^\bullet) \to 0. \tag{2.2}$$

The σ-truncations satisfy:

$$H^i\big(\sigma_{>n}(X^\bullet)\big) = \begin{cases} H^i(X^\bullet) & i > n, \\ 0 & i \le n; \end{cases} \qquad H^i\big(\sigma_{\le n}(X^\bullet)\big) = \begin{cases} H^i(X^\bullet) & i \le n, \\ 0 & i > n. \end{cases}$$

Lemma 6. *For any complex $X^\bullet$ over $\mathcal{A}$, there exist triangles in* $\mathrm{D}(\mathcal{A})$*:*

$$\tau_{>n}(X^\bullet) \to \tau_{\ge n}(X^\bullet) \to X^n \to T\tau_{>n}(X^\bullet), \tag{2.3}$$
$$H^n(X^\bullet) \to \sigma_{\ge n}(X^\bullet) \to \sigma_{>n}(X^\bullet) \to TH^n(X^\bullet). \tag{2.4}$$

Proof. Triangle (3) comes from the short exact sequence

$$0 \to \tau_{>n}(X^\bullet) \to \tau_{\ge n}(X^\bullet) \to X^n \to 0.$$

For (4), define $\sigma'_{\ge n}(X^\bullet)$ by

$$\cdots \to 0 \to X^n / \operatorname{im} d^{n-1} \to X^{n+1} \to \cdots .$$

There is a quasi-isomorphism $\sigma_{\ge n}(X^\bullet) \to \sigma'_{\ge n}(X^\bullet)$, and an exact sequence

$$0 \to H^n(X^\bullet) \to \sigma'_{\ge n}(X^\bullet) \to \sigma_{>n}(X^\bullet) \to 0.$$

This induces the required triangle in $\mathrm{D}(\mathcal{A})$. □

Proof of Proposition 1.7.1. (i) For $X^\bullet \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$, use descending induction on n to show $\eta\big(\sigma_{>n}(X^\bullet)\big)$ is an isomorphism. For large n, $\sigma_{>n}(X^\bullet)$ is acyclic, hence trivial. The induction step follows from triangle (4) and Proposition 1.1(c). For small n, $X^\bullet \cong \sigma_{>n}(X^\bullet)$ in $\mathrm{D}(\mathcal{A})$, done.

(ii) Let $X^\bullet \in \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$. It suffices to show $H^j(\eta(X^\bullet))$ is an isomorphism for all j. Given j, pick $n_1 \ge j+2$, choose n_2 from the way-out property for F, G. Since $H^i(\sigma_{>n_2}(X^\bullet)) = 0$ for $i \le n_2$, we have $H^i(F(\sigma_{>n_2}(X^\bullet))) = H^i(G(\sigma_{>n_2}(X^\bullet))) = 0$ for $i < n_1$, in particular for $i = j, j+1$.

By the long exact cohomology sequence:

$$H^j\big(F(\sigma_{\le n_2}(X^\bullet))\big) \xrightarrow{\sim} H^j\big(F(X^\bullet)\big),$$
$$H^j\big(G(\sigma_{\le n_2}(X^\bullet))\big) \xrightarrow{\sim} H^j\big(G(X^\bullet)\big).$$

But $\sigma_{\leq n_2}(X^\bullet) \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$, so η is an isomorphism there. Hence $H^j(\eta(X^\bullet))$ is an isomorphism.

(iii) Decompose $X^\bullet$ via $\sigma_{\leq 0}, \sigma_{>0}$, apply (ii) to each piece, then glue via the triangle from (2).

(iv) Use Lemma 1.4.6 to resolve any $X \in \mathcal{A}'$ by a complex $I^\bullet$ with terms in P. Apply (i),(ii) with τ-truncations instead of σ-truncations. □

Proposition 15. *Let $\mathcal{A}' \subseteq \mathcal{A}$, $\mathcal{B}' \subseteq \mathcal{B}$ be thick abelian categories, $F\colon \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ a triangulated functor.*

(i) If $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in \operatorname{Ob}\mathcal{A}'$, then $F(X^\bullet) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X^\bullet \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$.

(ii) If $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in \operatorname{Ob}\mathcal{A}'$ and F is way-out right, then $F(X^\bullet) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X^\bullet \in \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$.

(iii) If F is way-out both directions, extend to all $X^\bullet \in \mathrm{D}_{\mathcal{A}'}(\mathcal{A})$.

(iv) Let $P \subseteq \operatorname{Ob}\mathcal{A}'$ such that every object of $\mathcal{A}'$ embeds into P. If $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in P$ and F is way-out right, then $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in \operatorname{Ob}\mathcal{A}'$.

Proof. Analogous to Proposition 1.7.1. □

Proposition 16. *Let $\mathcal{A}$ have enough injectives, $F\colon \mathcal{A} \to \mathcal{B}$ additive of cohomological dimension $\leq n$. Let $P \subseteq \operatorname{Ob}\mathcal{A}$ be all X with $R^iF(X) = 0$ for $i \neq n$. Assume every object of $\mathcal{A}$ is a quotient of an object in P. Set $G = R^nF$. Then $\mathbf{R}F, \mathbf{L}G$ exist, and there is a functorial isomorphism*

$$\psi\colon \mathbf{R}F \xrightarrow{\sim} \mathbf{L}G[-n],$$

where $[-n]$ denotes shifting right by n degrees.

Proof. Existence of $\mathbf{R}F$ follows from Corollary 1.5.3. Existence of $\mathbf{L}G$ follows by dualizing Theorem 1.5.1 for left derived functors.

The class P satisfies the dual conditions of Lemma 1.4.6 and Corollary 1.5.3. Let $\mathrm{L} \subseteq \mathrm{K}(\mathcal{A})$ be complexes of objects in P; the hypotheses of Theorem 1.5.1 hold, so $\mathbf{L}G$ exists.

Since $\mathrm{L}_{\mathrm{Qis}} \to \mathrm{D}(\mathcal{A})$ is an equivalence, it suffices to define ψ on complexes in L: for any $X^\bullet \in \mathrm{L}$,

$$\psi(X^\bullet)\colon \mathbf{R}F(X^\bullet) \to \mathbf{L}G(X^\bullet)[-n],$$

compatible with morphisms of complexes.

Given $X^\bullet \in \mathrm{L}$, choose a Cartan–Eilenberg resolution $C^{\cdot\cdot}$. Define the truncated double complex $C'^{\cdot\cdot} = \sigma^{\mathrm{II}}_{\leq n}(C^{\cdot\cdot})$ by

$$C'^{p,q} = \begin{cases} C^{p,q} & q < n, \\ \ker d_2^{p,n} & q = n, \\ 0 & q > n. \end{cases}$$

For each $p \in \mathbb{Z}$, we have an exact sequence

$$0 \to X^p \to C'^{p,0} \to C'^{p,1} \to \cdots \to C'^{p,n-1} \to C'^{p,n} \to 0.$$

The terms $C'^{p,0}, \ldots, C'^{p,n-1}$ are injective; $X^p \in P$ implies $C'^{p,n}$ is F-acyclic. This resolution computes $\mathbf{R}F(X^p)$, giving an isomorphism

$$\varphi(t^p)\colon G(X^p) = R^nF(X^p) \to F(C'^{p,n}) \big/ \operatorname{im} F(C'^{p,n-1}),$$

where $t^p\colon X^p \to C^{p,\bullet}$ is the resolution map. This induces a morphism $\alpha^p\colon F(C^{p,n}) \to G(X^p)$, and hence a double complex morphism

$$\alpha\colon F(C'^{\cdot\cdot}) \to G(X^\bullet),$$

with $G(X^\bullet)$ concentrated in the n-th row.

Passing to associated simple complexes:

$$s(\alpha)\colon F\big(s(C'^{\cdot\cdot})\big) \to G(X^\bullet)[-n].$$

By Lemma 1.7.5(f) below, the augmentation $u\colon X^\bullet \to s(C^{\cdot\cdot})$ is a quasi-isomorphism into an F-acyclic complex, yielding an isomorphism

$$\varphi(u)\colon \mathbf{R}F(X^\bullet) \to F\big(s(C^{\cdot\cdot})\big).$$

Since $X^\bullet$ consists of G-acyclic objects, we have

$$\varphi(\mathrm{id}_{X^\bullet})\colon G(X^\bullet) \to \mathbf{L}G(X^\bullet).$$

Composing these gives the desired morphism

$$\psi(X^\bullet)\colon \mathbf{R}F(X^\bullet) \to \mathbf{L}G(X^\bullet)[-n].$$

The morphism $\psi(X^\bullet)$ is independent of the Cartan–Eilenberg resolution choice. To see ψ is an isomorphism: $\mathbf{R}F, \mathbf{L}G$ are way-out in both directions. By the way-out functor lemma (Proposition 1.7.1), reduce to checking $\psi(X)$ is an isomorphism for $X \in P$, which holds by construction. $\square$

Proposition 17. *Let $\mathcal{A}$ have enough injectives, $X^\bullet \in \mathrm{K}^+(\mathcal{A})$. The following are equivalent:*

(i) *$X^\bullet$ admits a quasi-isomorphism into a bounded complex of injective objects of $\mathcal{A}$.*

(ii) *The functor $\mathbf{R}\mathcal{H}om(\,\cdot\,, X^\bullet)\colon \mathrm{D}(\mathcal{A})^\circ \to \mathrm{D}(\mathbf{Ab})$ is way-out left (hence way-out both ways).*

(iii) *There exists $n_0 \in \mathbb{Z}$ such that $\mathrm{Ext}^i(Y, X^\bullet) = 0$ for all $Y \in \mathrm{Ob}\,\mathcal{A}$ and all $i > n_0$.*

Proof. *(i)* $\Rightarrow$ *(ii)*: Assume $X^\bullet$ is bounded injective. Then $\mathbf{R}\mathcal{H}om(Y^\bullet, X^\bullet) \cong \mathcal{H}om(Y^\bullet, X^\bullet)$, which is clearly way-out left.

(ii) $\Rightarrow$ *(iii)*: Since $F = \mathbf{R}\mathcal{H}om(\,\cdot\,, X^\bullet)$ is way-out left, choose n_0 such that $H^i(F(Y^\bullet)) = 0$ whenever $H^i(Y^\bullet) = 0$ for $i < n_0$. For fixed $Y \in \mathcal{A}$, set $Y'^\bullet = T^{-n_0}(Y)$. Then $H^i(Y'^\bullet) = 0$ for $i < n_0$, so $H^i(F(Y'^\bullet)) = 0$ for $i > 0$. But

$$H^i(F(Y'^\bullet)) = H^i\big(F(T^{-n_0}Y)\big) = H^{i+n_0}(F(Y)),$$

hence $\mathrm{Ext}^{i+n_0}(Y, X^\bullet) = 0$ for all $i > 0$, i.e., $\mathrm{Ext}^j(Y, X^\bullet) = 0$ for $j > n_0$.

(iii) $\Rightarrow$ *(i)*: Take an injective resolution $X'^\bullet \in \mathrm{K}^+(\mathcal{A})$ of $X^\bullet$ (Lemma 1.4.6). Suppose $H^m(X'^\bullet) \neq 0$ for some $m > n_0$. From the exact sequence

$$0 \to B^m(X'^\bullet) \to Z^m(X'^\bullet) \to H^m(X'^\bullet) \to 0,$$

the inclusion is proper. Choose $Y = Z^m(X'^\bullet)$; then

$$\mathrm{Hom}(Y, B^m(X'^\bullet)) \subsetneq \mathrm{Hom}(Y, Z^m(X'^\bullet)).$$

But $\mathrm{Ext}^m(Y, X^\bullet) = 0$ by hypothesis, which forces the map to be an isomorphism, a contradiction. Thus $H^i(X'^\bullet) = 0$ for all $i > n_0$.

For $n \geq n_0$, the truncation map $i\colon \sigma_{\leq n}(X'^\bullet) \to X'^\bullet$ is a quasi-isomorphism. For $n \geq n_0$, $Z^{n+1}(X'^\bullet) = B^{n+1}(X'^\bullet)$ is injective. Consider the exact sequence

$$0 \to \sigma_{\leq n}(X'^\bullet) \to \tau_{\leq n}(X'^\bullet) \to B^{n+1}(X'^\bullet)[-n] \to 0.$$

We show $\mathrm{Ext}^1(Y, B^{n+1}) = 0$ for all $Y \in \mathcal{A}$. We have

$$\mathrm{Ext}^1(Y, B^{n+1}) = \mathrm{Ext}^{n+1}\left(Y, B^{n+1}[-n]\right).$$

From the Ext long exact sequence:

$$\cdots \to \mathrm{Ext}^{n+1}\left(Y, \tau_{\leq n}(X'^\bullet)\right) \to \mathrm{Ext}^{n+1}\left(Y, B^{n+1}[-n]\right) \to \mathrm{Ext}^{n+2}\left(Y, \sigma_{\leq n}(X'^\bullet)\right) \to \cdots.$$

The left term vanishes because $\tau_{\leq n}(X'^\bullet)$ is bounded injective; the right term vanishes by hypothesis. Hence $\mathrm{Ext}^1(Y, B^{n+1}) = 0$, so B^{n+1} is injective. Thus $\sigma_{\leq n}(X'^\bullet)$ is a bounded injective complex quasi-isomorphic to $X^\bullet$. $\square$

Definition 16. *If* $X^\bullet \in \mathrm{K}^+(\mathcal{A})$ *satisfies the equivalent conditions of Proposition 1.7.6, we say* $X^\bullet$ *has* finite injective dimension*.*

Corollary 4. *The complexes of finite injective dimension form a triangulated localizing subcategory of* $\mathrm{K}(\mathcal{A})$*.*

Proof. Closure under triangles follows from the Ext long exact sequence. Localizing follows from Proposition 1.3.3: any complex quasi-isomorphic to a finite injective dimension complex also has finite injective dimension. □

Definition 17. *Denote by* $\mathrm{K}^+(\mathcal{A})_{\mathrm{fid}}$ *the full subcategory of* $\mathrm{K}^+(\mathcal{A})$ *consisting of complexes of finite injective dimension. The corresponding derived subcategory is* $\mathrm{D}^+(\mathcal{A})_{\mathrm{fid}}$*.*

Remark. These subcategories differ from $\mathrm{K}_{\mathcal{A}'}(\mathcal{A}), \mathrm{D}_{\mathcal{A}'}(\mathcal{A})$, as they cannot be characterized solely by conditions on the cohomology objects $H^i(X^\bullet)$.

Chapter 3

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Index of Notations

Of course there is the usual collection of variable notations: A, B for abelian categories or rings, functors or sheaves, X, Y for preschemes, x, y for points of them, f, g for morphisms of preschemes, ϕ for morphisms of functors on sheaves, etc.

In general no underline denotes a group, as $\mathrm{Hom}(\mathcal{F}, \mathcal{G})$, one underline denotes a sheaf, as $\underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$ or $\underline{\mathrm{Tor}}(\mathcal{F}, \mathcal{G})$, two underlines denotes an object in a derived category, as $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}^{\bullet}(\mathcal{F}^{\bullet}, \mathcal{G}^{\bullet})$, and a dot denotes a complex.

In the list of stable notations below, we have distinguished five categories: Latin alphabet, other alphabets, superscripts, subscripts, and arbitrary symbols.

Latin alphabet

Symbol	Meaning	Reference
Ab	the category of abelian groups	
acc	ascending chain condition	V.2
Ass	associated primes	
B	boundaries of a complex	
Coh	category of coherent sheaves	II.1
Coz	category of Cousin complexes	
D	dualizing functor	V.0
$\mathcal{D}(\mathcal{A})$	derived category	I.4
$\mathcal{D}(X)$	$= \mathcal{D}(\mathrm{Mod}(X))$	II.1
dcc	descending chain condition	V.2
dim	dimension	
Dual	category of dualizing complexes	VI.1
E	associated Cousin complex	IV.2, IV.3
Ext	Ext groups	I.6
$\underline{\mathrm{Ext}}$	sheaf Ext	II.3

Symbol	Meaning	Reference
fid	finite injective dimension	I.7.6, II.7.20
FR1–FR5	axioms of multiplicative systems	I.3
H	cohomology	I.1, IV.1
Hom	Hom groups	I.6
$\underline{\mathrm{Hom}}$	sheaf Hom	II.3
i_x	closed immersion	II.7
Icz	category of injective Cousin complexes	
id	identity map	
im	image	
$J(x), J(x, x')$	standard injectives	II.7
$\mathcal{K}(A)$		II.7
$k(x)$	residue field of a point x	
$K_\bullet(\underline{f}), K^\bullet(\underline{f}; F)$	Koszul complexes	III.7
ker	kernel	
L1–L3	axioms of direct systems	I.3
$DerL$	left derived functor	I.5
lffr	locally free of finite rank	II.5
Lno	category of locally noetherian preschemes	III.8
Mod(X)	category of $\mathcal{O}_X$-modules	II.1
Ob	objects of a category	
pfid	pointwise finite injective dimension	V.8
Ptwdual	category of pointwise dualizing complexes	VI.1
Q	functor into derived category	I.3
Qco	category of quasi-coherent sheaves	II.1
Qis	quasi-isomorphisms	I.4
$\underline{\underline{\mathrm{R}}}$	right derived functor	I.5
Res	category of residual complexes	VI.1
	residue symbol	
	residue of a differential	III.9
Spec	spectrum of a ring	VII.1
Supp	support of a sheaf or module	II.7
T	translation functor	I.1
$\underline{\mathrm{Tor}}$		
Tr	trace map	III.10, VI.4, VII.1
TR1–TR4	axioms of a triangulated category	I.1
TRA	axioms of trace	III.10.5, VI.4.2

Symbol	Meaning	Reference
tr. d	transcendence degree	
$\operatorname{Tr} f$	trace for a finite morphism	III.6
$\operatorname{Tr} p$	trace for a projective morphism	III.11
VAR	axioms for $f^!$ or f^Δ	III.8.7, VI.3.1
Z	cycles of a complex	

Greek and other alphabets

Symbol	Meaning	Reference
γ	trace for smooth morphism	III.3.4, III.11.2, VII.4.1
Γ	global sections of a sheaf	II.1.5
ζ	change of differentials	III.7.3
η	fundamental local isomorphism	III.7.3
θ	duality morphism	III.7.3
ξ	map of a functor to its derived functor	III.7.3
$\sigma_{\geq n}$	truncated complex	I.7
$\tau_{\geq n}$	truncated complex	I.7
ψ	residue isomorphism	III.8
ω	sheaf of differentials	III.1
Ω	sheaf of differentials	III.1
$\mathcal{O}_X$	structure sheaf of a prescheme	
$\mathfrak{m}$	maximal ideal of a ring	
$\mathfrak{p}$	prime ideal of a ring	
$\mathbb{A}$	affine space	
$\mathbb{P}$	projective space	
$\mathbb{Q}$	rational numbers	
$\mathbb{Z}$	integers	

Subscripts

Symbol	Meaning	Reference

$\mathcal{A}'$	as in $\mathcal{K}_{\mathcal{A}'}(\mathcal{A}), \mathcal{D}_{\mathcal{A}'}(\mathcal{A})$	I.4
c	as in $\mathcal{D}_c(X)$	II.1
fid	as in $\mathcal{D}(X)_{\mathrm{fid}}$	I.7
fTd	as in $\mathcal{D}(X)_{\mathrm{fTd}}$	II.4
$\mathrm{Gor}(Z^\bullet)$	as in $\mathcal{D}(X)_{\mathrm{Gor}(Z^\bullet)}$	IV.3
$\mathfrak{p}$	as in $A_\mathfrak{p}$, $M_\mathfrak{p}$: localization at a prime ideal	
qc	as in $\mathcal{D}_{\mathrm{qc}}(X)$	II.1
x	as in $\mathcal{O}_x$: local ring at a point x	
	as in $\mathcal{F}_x$: stalk of a sheaf at x	
	as in H^i_x: local cohomology	
Z	as in Γ_Z, H^i_Z	IV.1
I, II	as in $\underline{\underline{\mathrm{R}}}_{\mathrm{I}}, \underline{\underline{\mathrm{R}}}_{\mathrm{II}}$	IV.1
$\geq n$	as in $\sigma_{\geq n}, \tau_{\geq n}$	IV.1
$*$	as in f_*: direct image	IV.1

Superscripts

Symbol	Meaning	Reference
b	as in $\mathcal{K}^b, \mathcal{D}^b$: bounded	I.2, I.4
i	as in X^i:i-th term of a complex	
	as in H^i:i cohomology group	
	as in R^iF:i-th derived functor	
y	as in f^y	VI.2
z	as in f^z	VI.2
$\bullet$	as in $\mathcal{F}^\bullet, X^\bullet$: denotes a complex	
$\bullet\bullet$	as in $\mathcal{C}^{\bullet\bullet}$: double complex	I.7
$\bullet$	as in $\mathcal{K}(A)^\bullet$: opposed category	
$*$	as in f^*: inverse image	I.7, II.4
$+$	as in $\mathcal{K}^+, \mathcal{D}^+, \underline{\underline{\mathrm{R}}}^+$: bounded below	I.2, I.4, I.5
$-$	as in $\mathcal{K}^-, \mathcal{D}^-, \underline{\underline{\mathrm{R}}}^-$: bounded above	I.2, I.4, I.5
!	as in $f^!$	Intr., III.8, VII.3
$\flat$	$f^\flat$	IX.6
$\sharp$	$f^\sharp$	VI.3
Δ	f^Δ	VI.3
$\vee$	as in $L^\vee$: dual sheaf	II.5.16, III.1
$-$	as in $\{x\}^-$: closure	

$\sim$	as in $\tilde{M}$: sheaf associated to a module
$\wedge$	as in $\hat{A}$: completion of local ring

Arbitrary Symbols

Symbol	Meaning	Reference
$\vert$	restriction	
$\gg$	enough larger than	
$\times$	product	
$\otimes$	tensor product	
$\oplus$	direct sum	
$\coprod$	disjoint union / sum	
$[\ ]$	shift operator	I.1
	closed interval	I.1
	reference to Bibliography or other chapter	
$\bigcup$	union	
$\{\ \}$	the set of	
$\to$	morphism	
$\mapsto$	effect of a map on elements	
$\cdot\to$	distinguished side of triangle	
$\dashrightarrow$	morphism being constructed	
$\circ$	composition of functors ($F \circ G$)	I.2

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Errata